

XQuality project

Comparison between Normal production & Ab-normal production

Case of the changes in the feedrate (80%)



Raw Data

Raw Data



Batch #1 & #2 :

- Part #7 to Part #60 : Normal process conditions
- Part #61 to Part #70 : Feedrate reduced to 80%
- Part #71 to Part #75 : 316L grade instead of 303 grade
- Part #76 to Part #77 : Normal process conditions for tool wear changes verification (if any)
- Part #78 to Part #81 : Tool references mistakes (voluntary)

Machining operations numbering



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Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

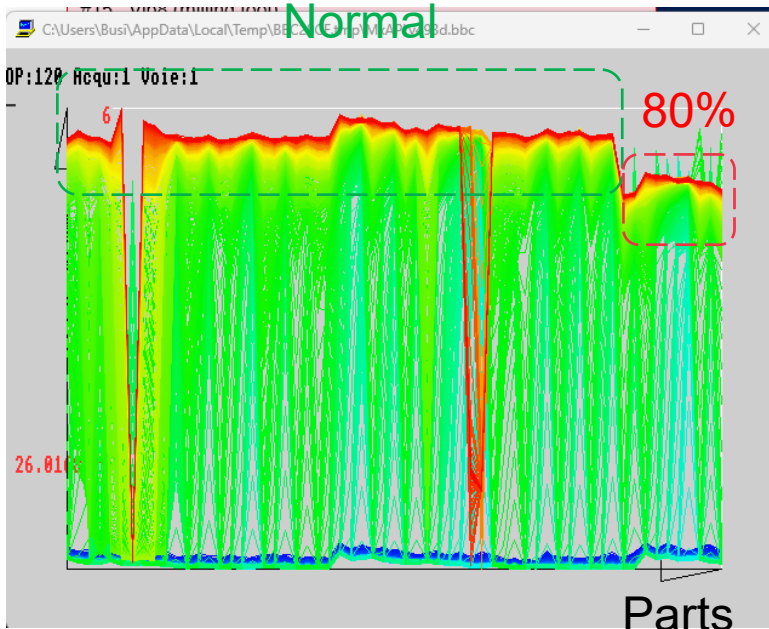
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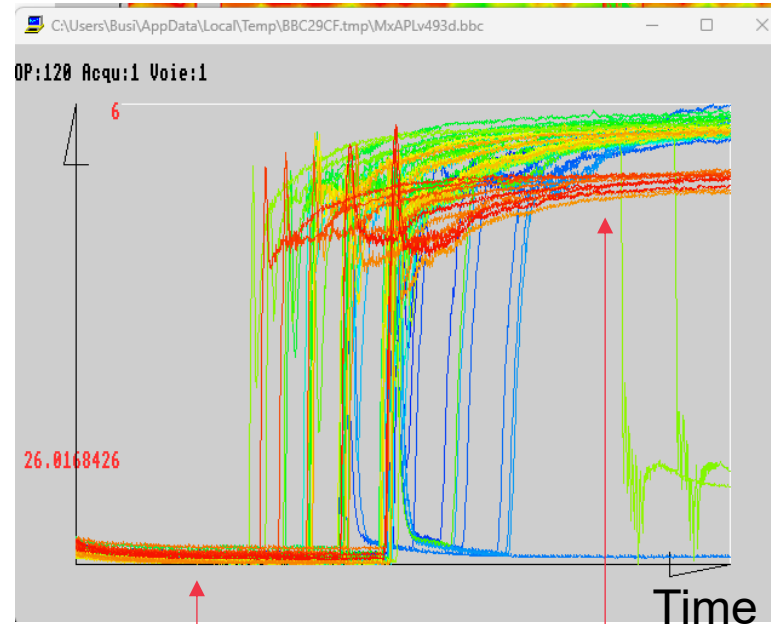
Machining OP 120

Spindle Power : Ai1

Power



Power

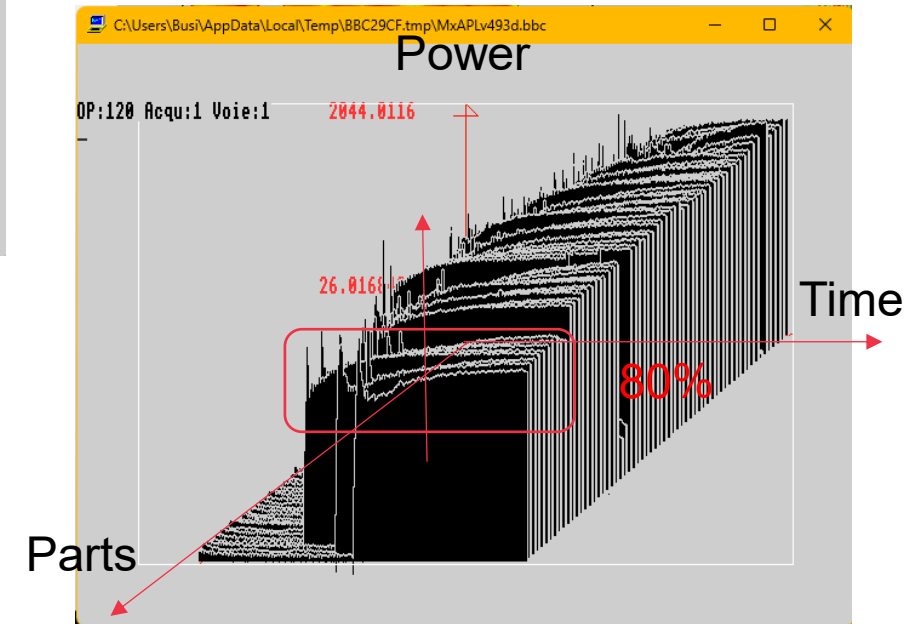
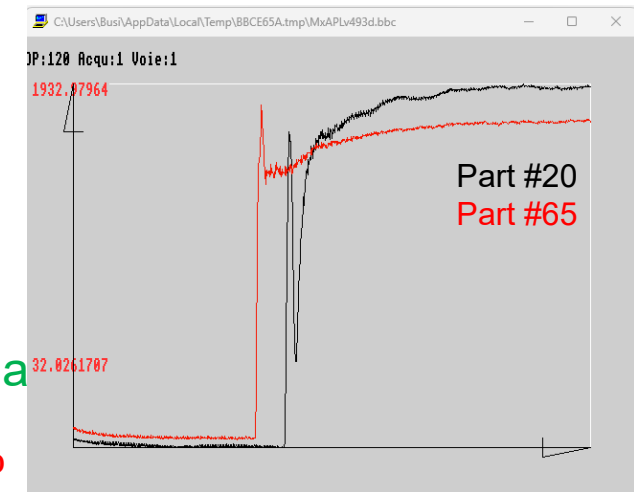


Before machining

In machining

Normal

80%



Rem : Temporal shifts sometimes !

Cutting time increases when feedrate decreases => OK

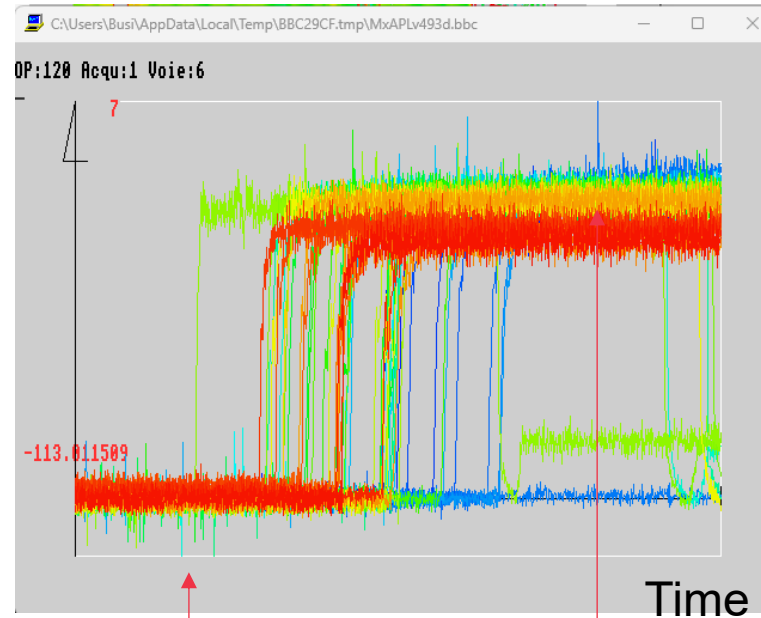
Spindle power decreases when feedrate decreases => OK

Cutting force : Fz1



Parts

Force

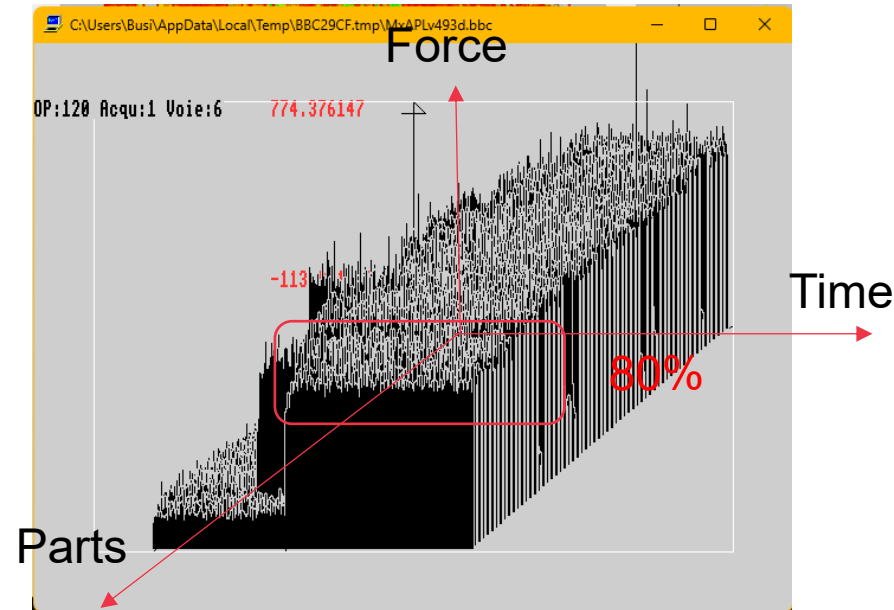
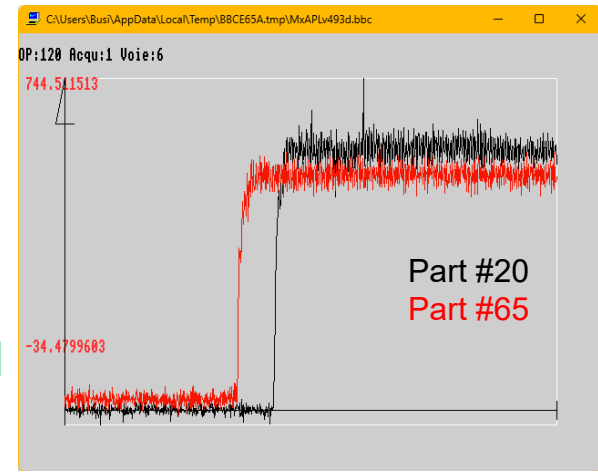


Before machining

In machining

Normal

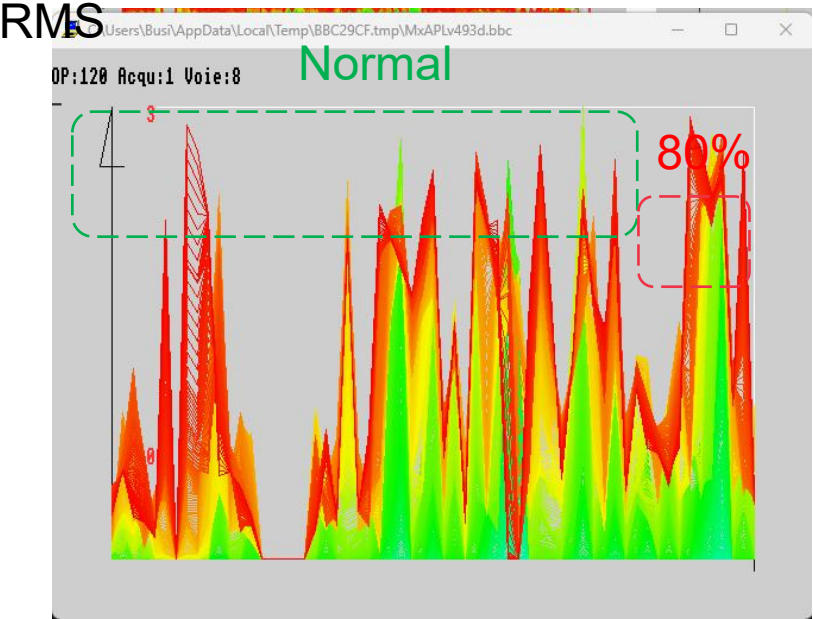
80%



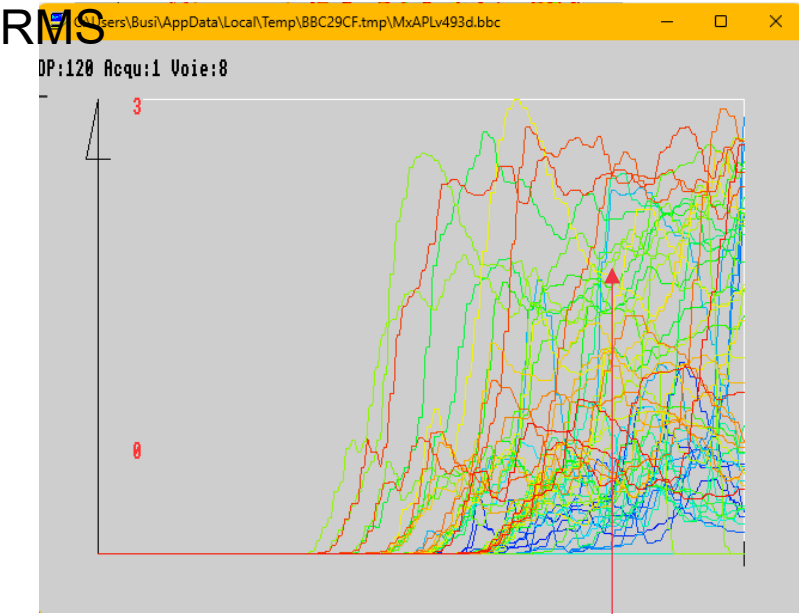
Rem : Temporal shifts sometimes !

Cutting Force decreases when feedrate decreases => OK

Vib : Vib1

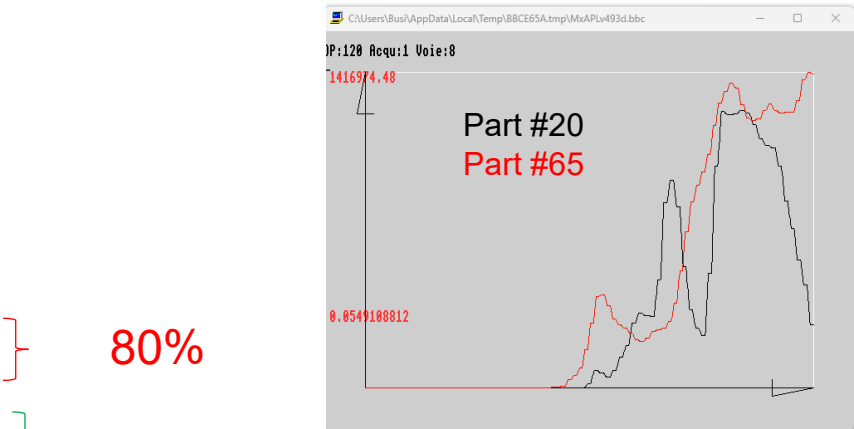


Parts



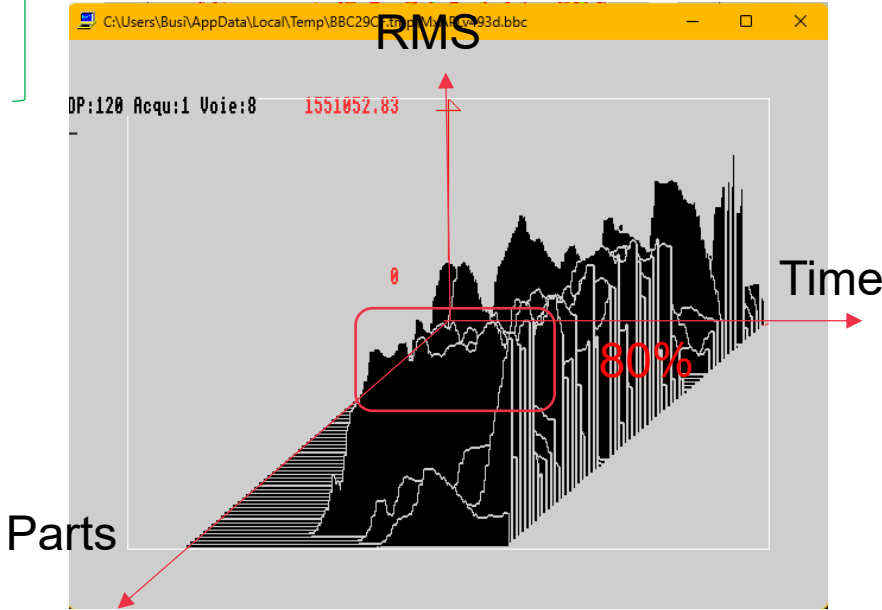
Before machining

In machining



80%

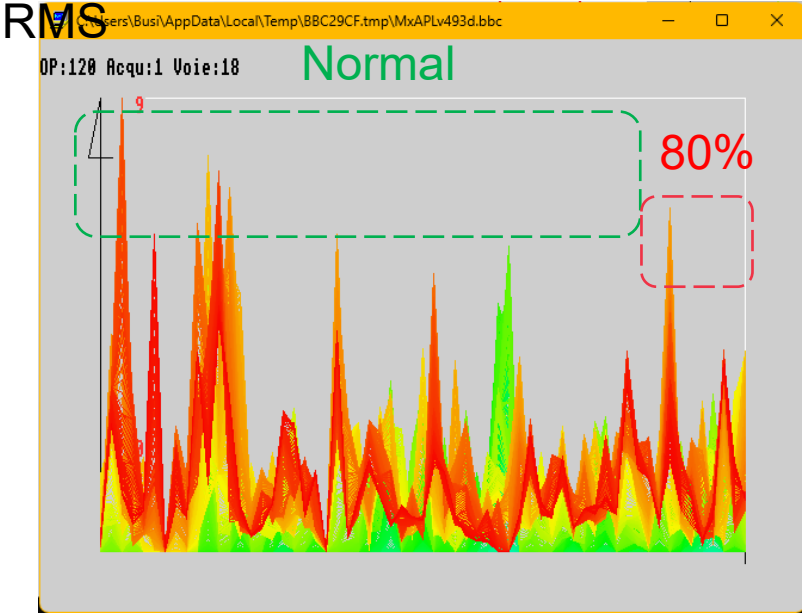
Normal



Parts

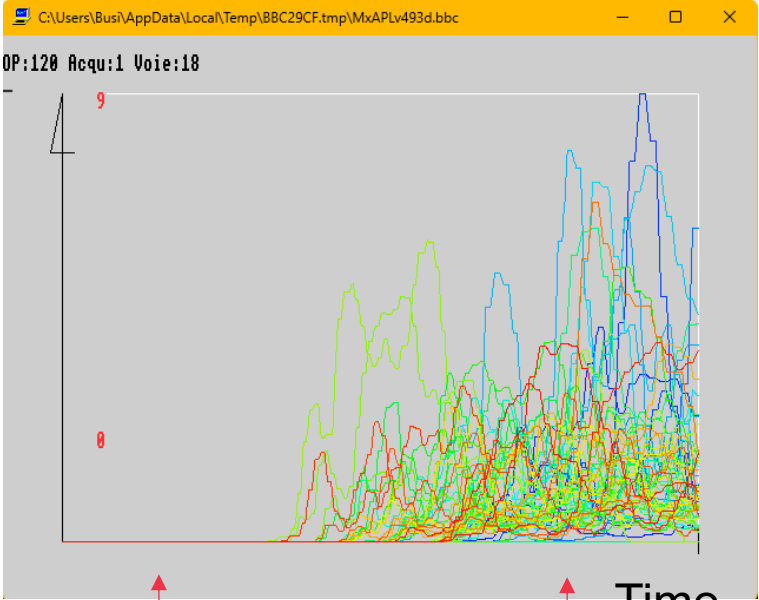
RMS remains the same when feedrate decreases

Audio : Vib 11



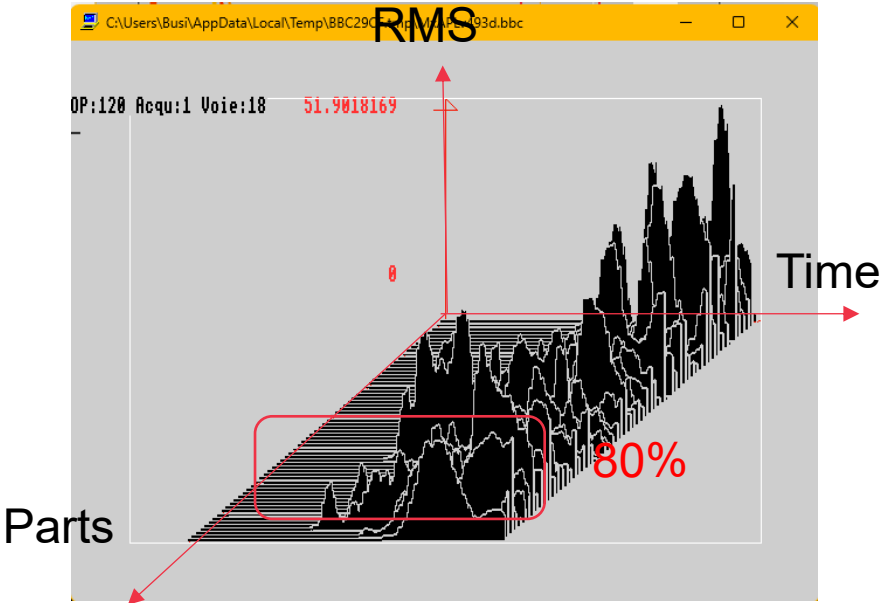
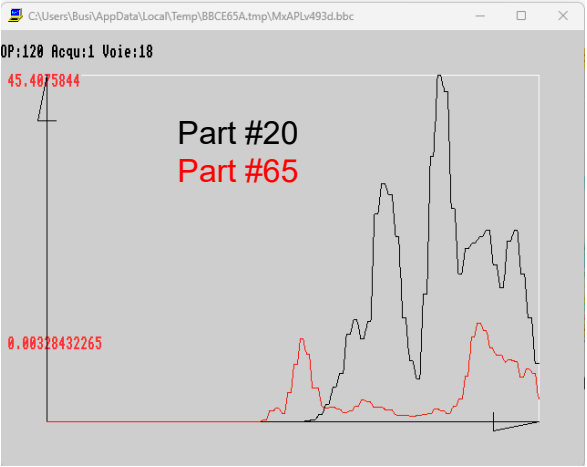
Parts

RMS



Before machining

In machining



Not obvious to see changes in Audio RMS signal

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- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

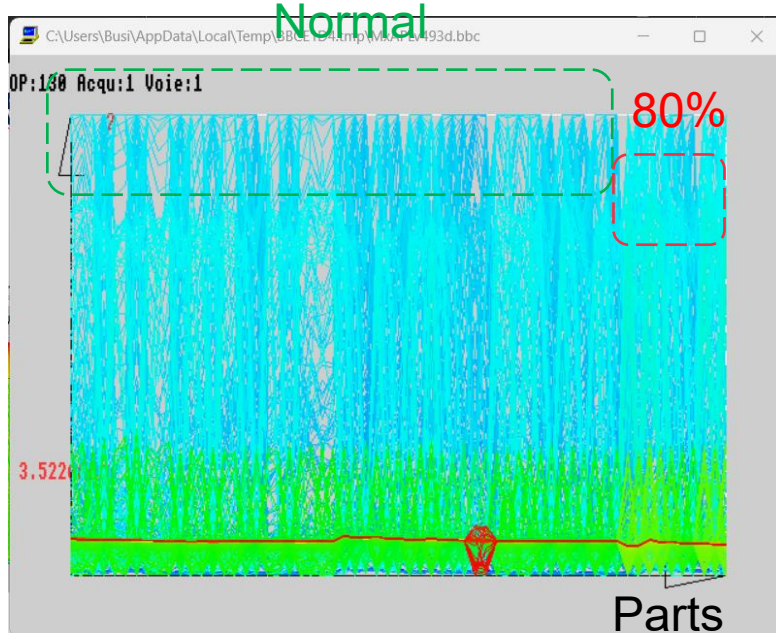
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Intermediate report Project n°316285 - Date 26/01/26 Page 91/188
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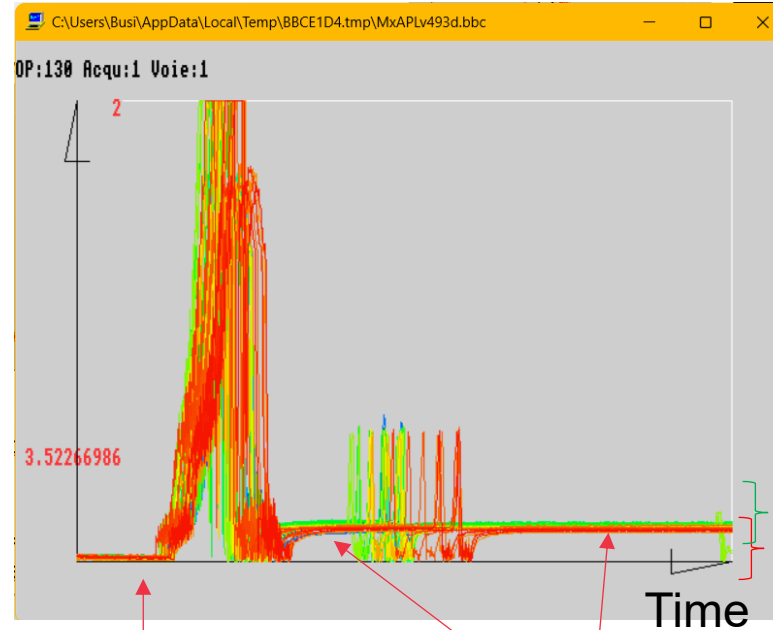
Machining OP 130

Spindle Power : Ai1

Power

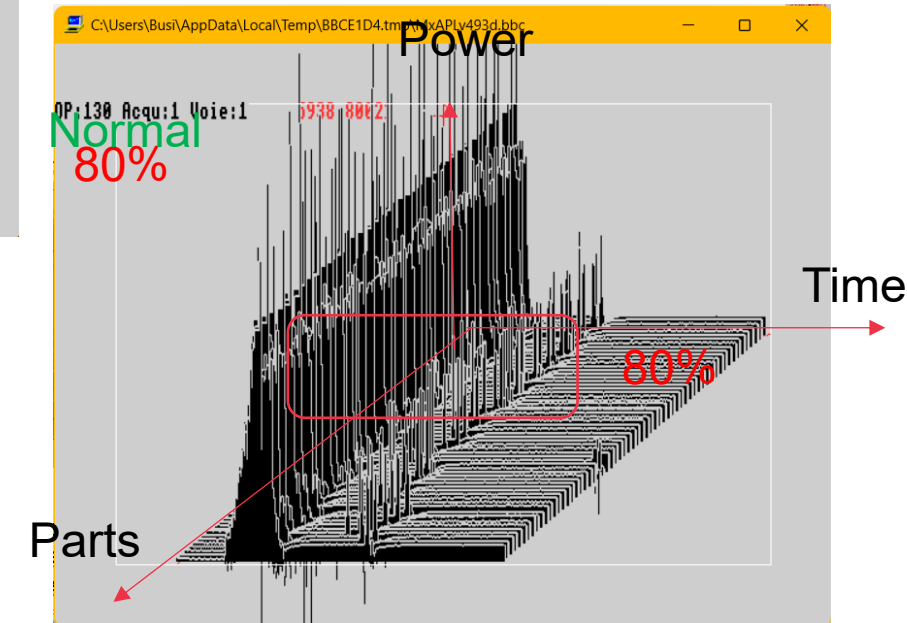
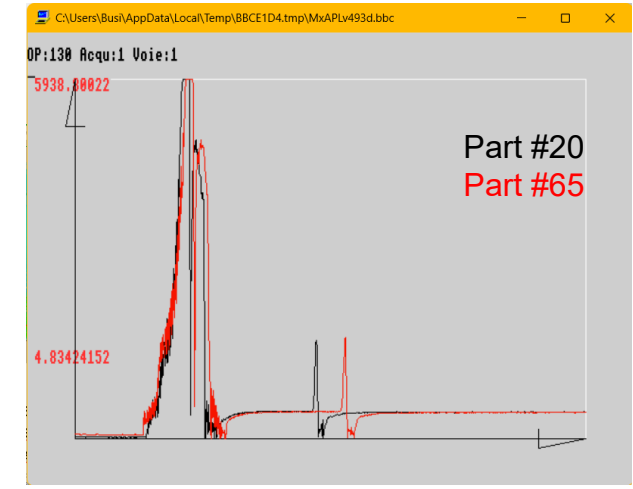


Power



Before machining

In machining



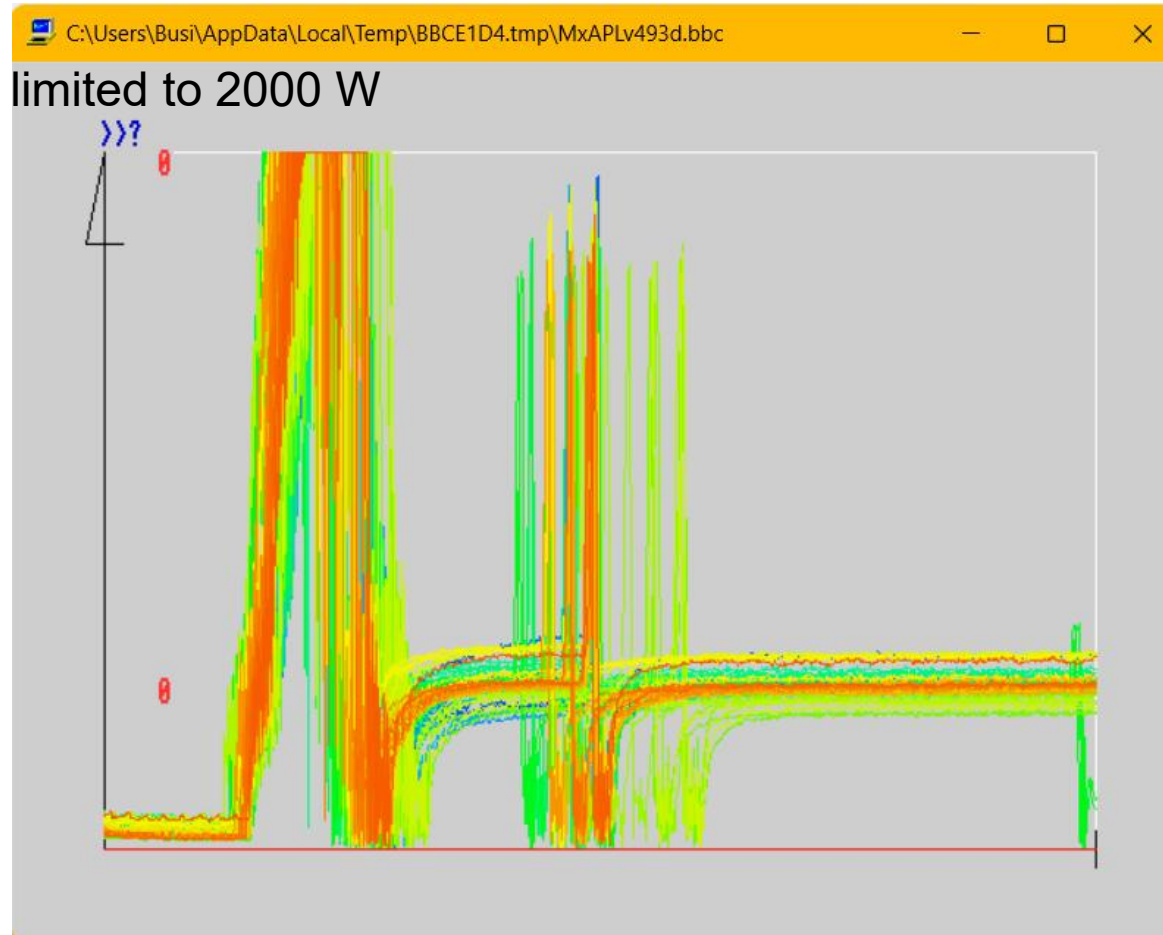
Rem : Temporal shifts sometimes !

Difficult to see here differences when en feedrate decreases => OK

Spindle Power : Ai1



Power limited to 2000 W

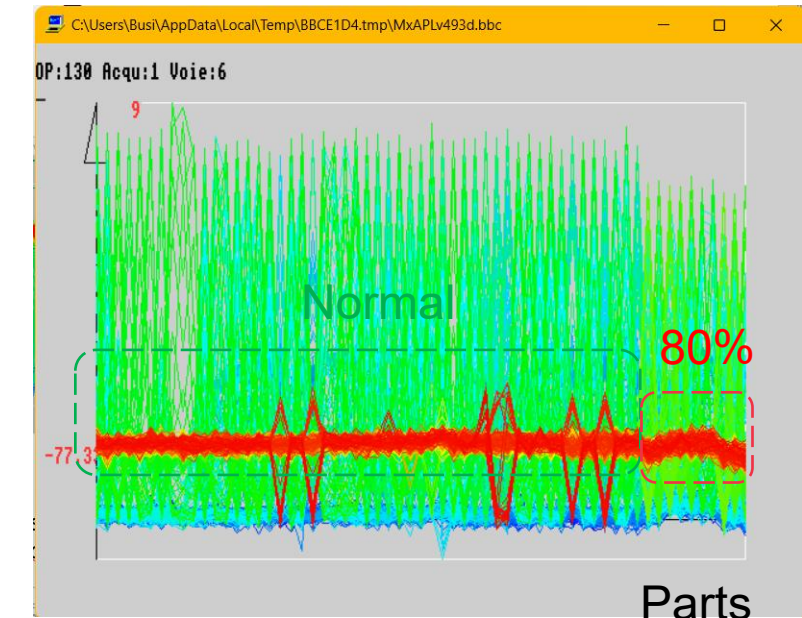


Rem : Temporal shifts sometimes !

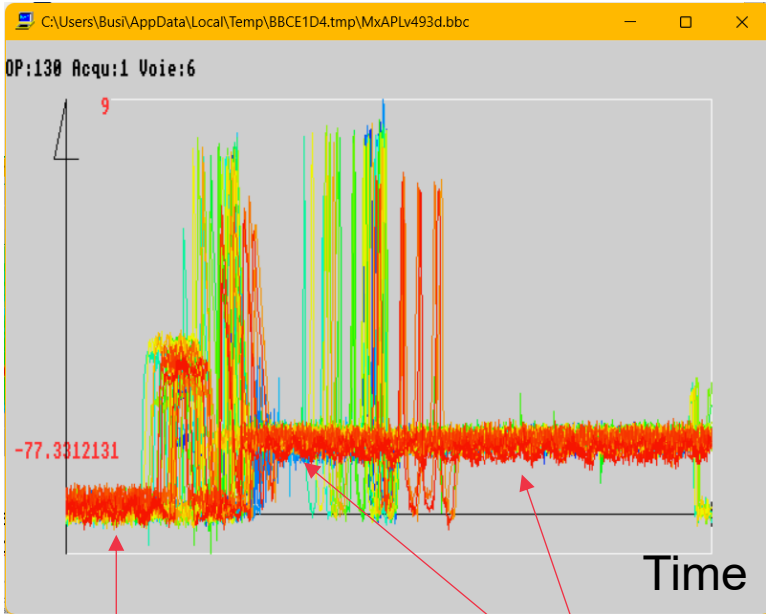
Even with a vertical zoom => Difficult to see here differences when feedrate decreases

Cutting force : Fz1

Force



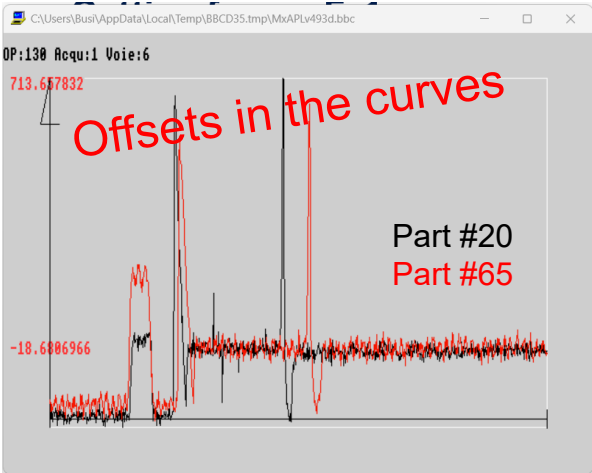
Force



Before machining

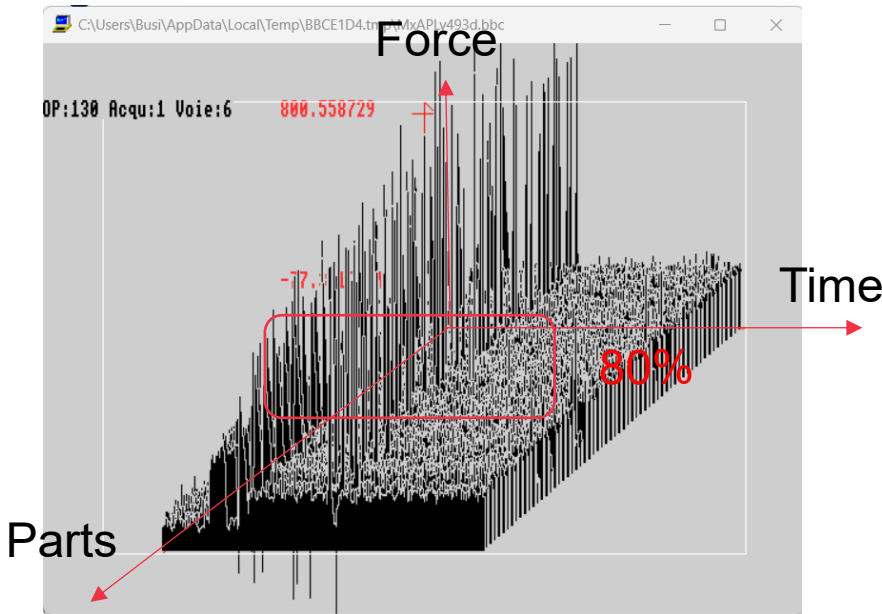
In machining

Normal
80%



Offsets in the curves

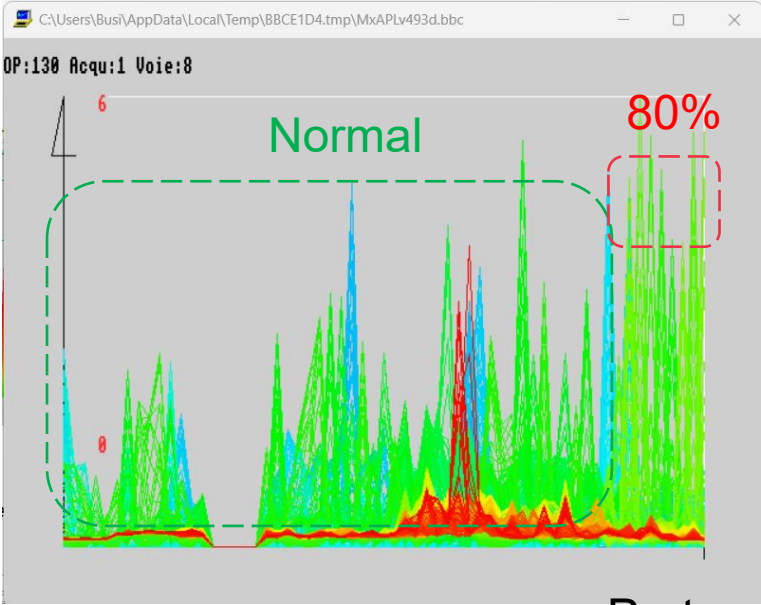
Part #20
Part #65



Rem : Temporal shifts sometimes !
Cutting Force decreases a little when feedrate decreases => OK

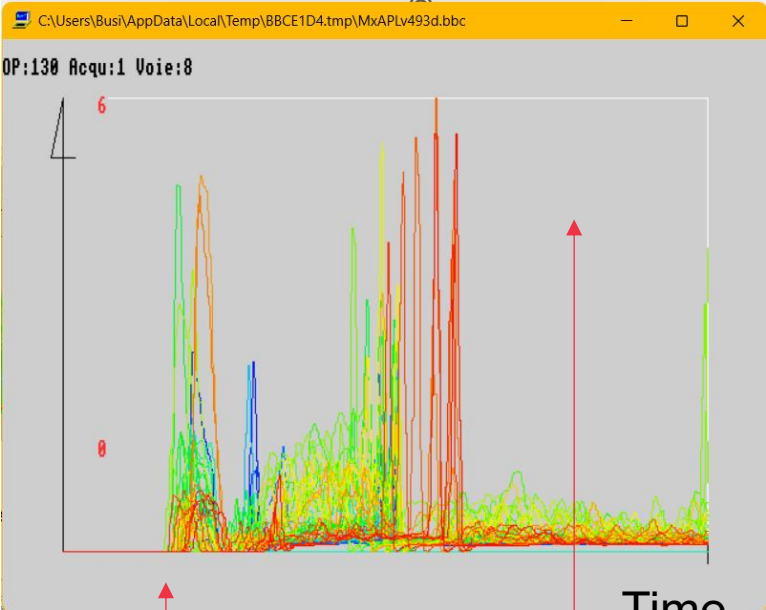
Vib : Vib1

RMS



Parts

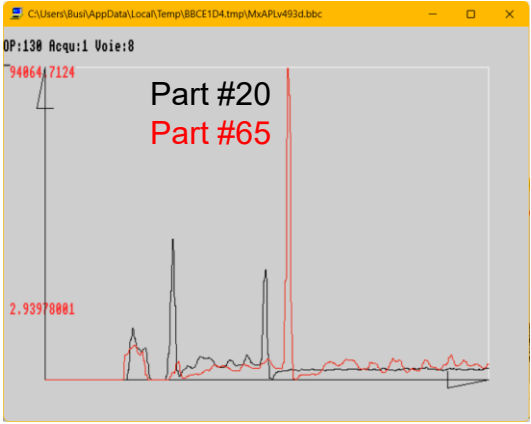
RMS



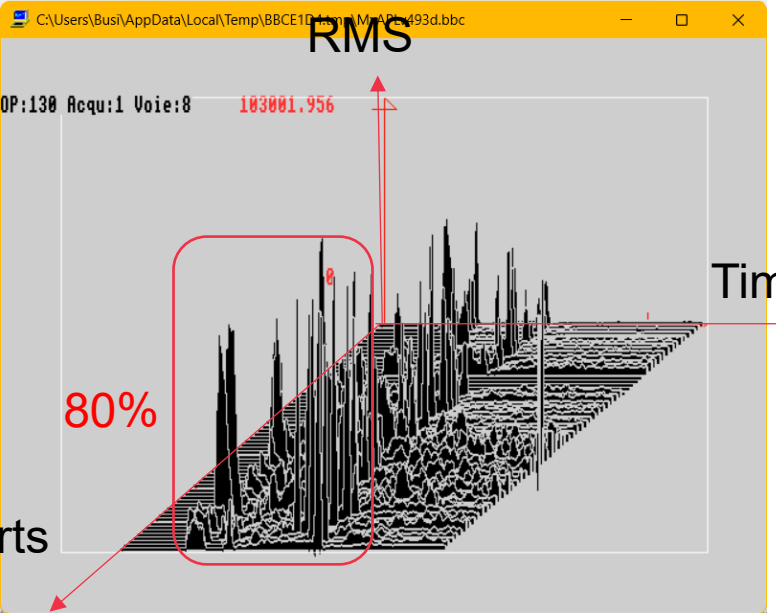
Before machining

In machining

80%



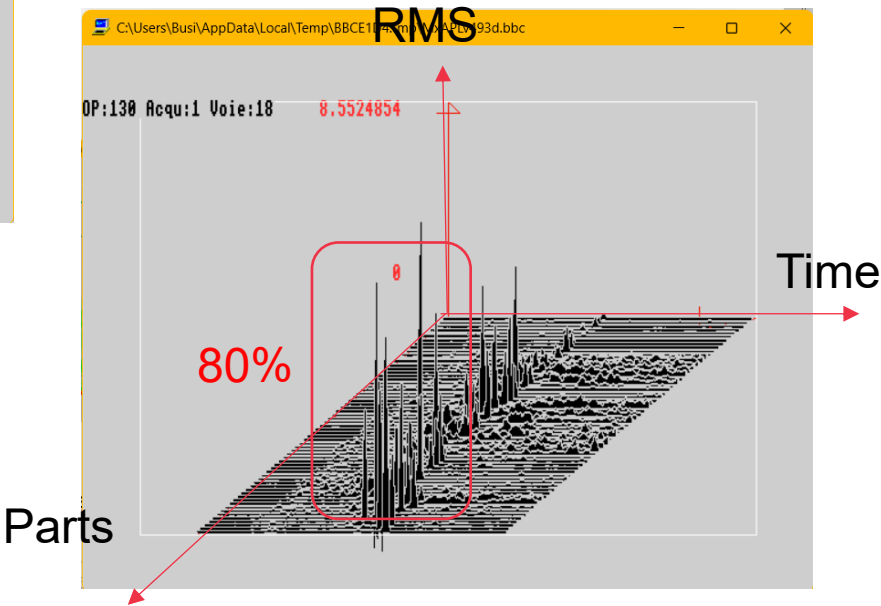
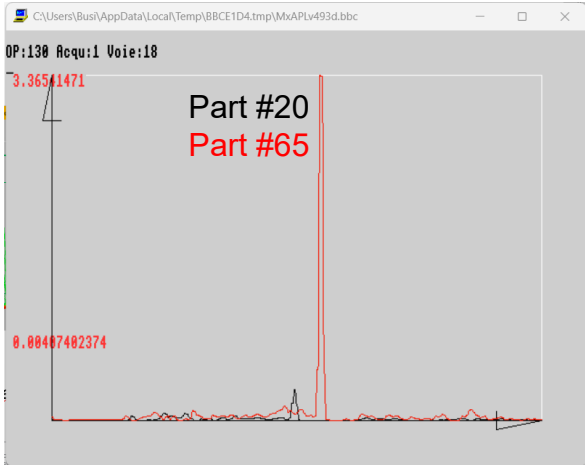
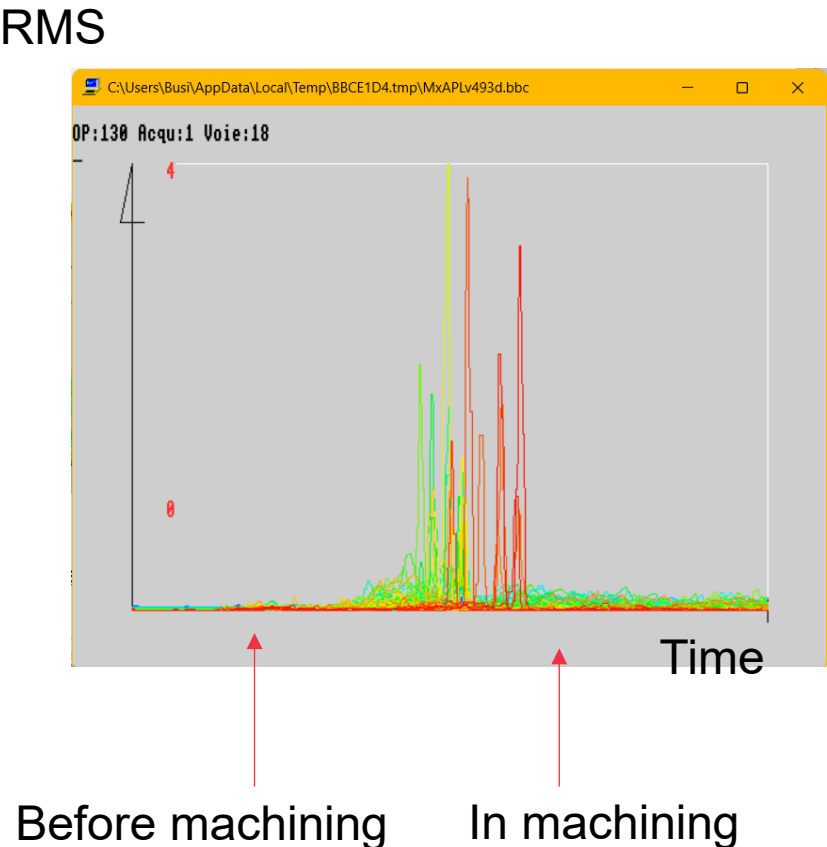
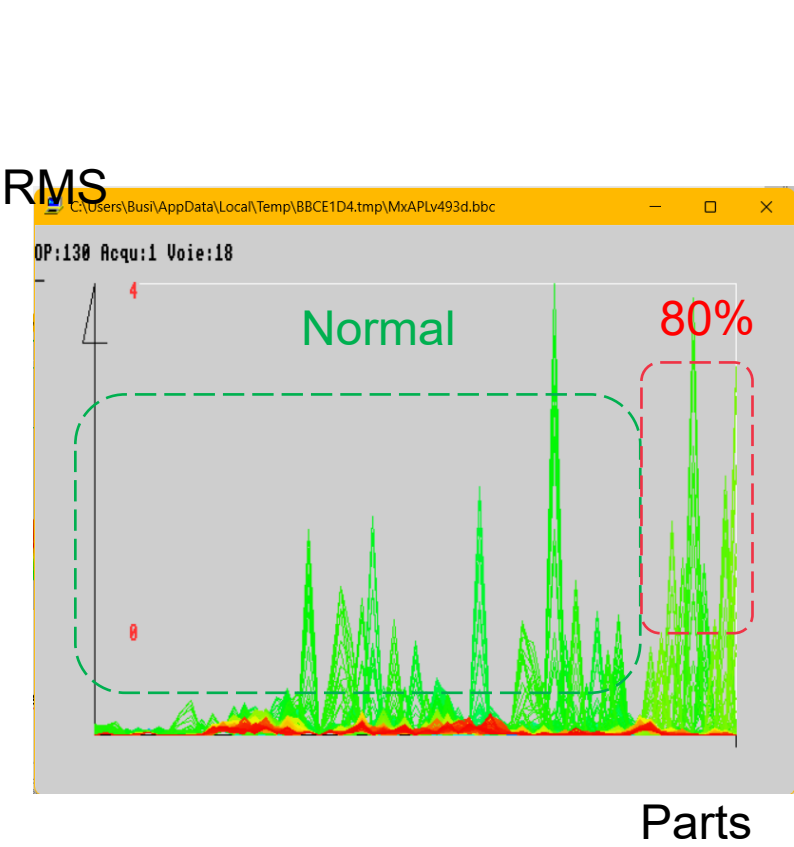
Normal



Parts

RMS increases (in the 2nd part of the signal) when feedrate decreases

Audio : Vib 11



RMS audio increases when feedrate decreases => OK

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

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#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

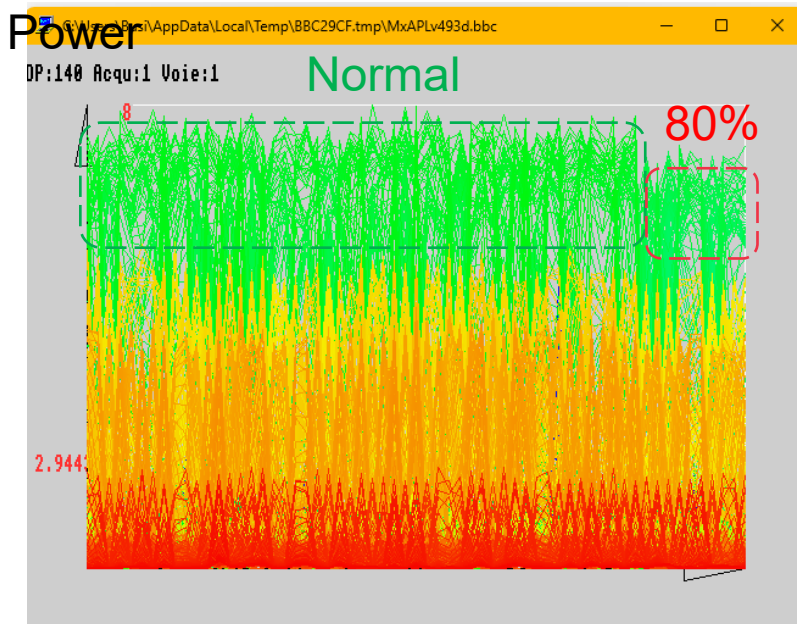
#18 : Vib11 (audio micro)

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Intermediate report Project n°316285 - Date 26/01/26 Page 91/188
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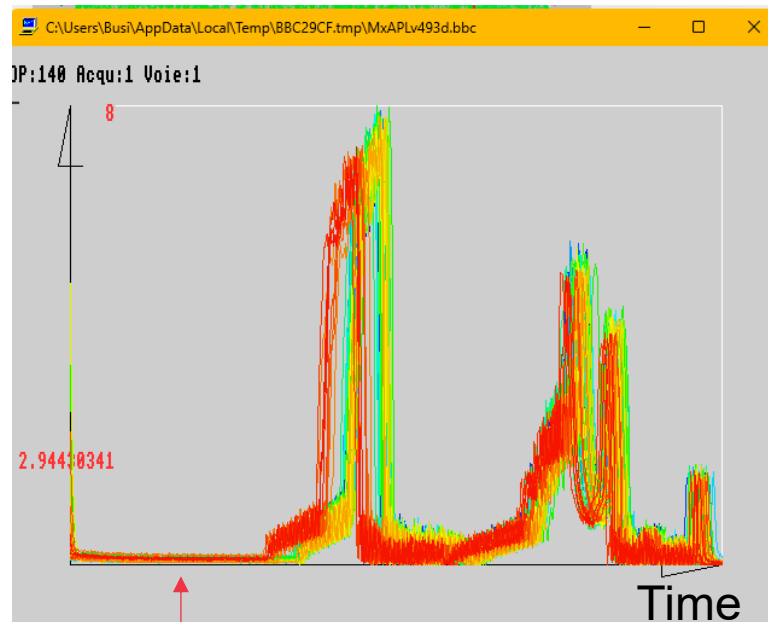
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<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 140

Spindle Power : Ai1



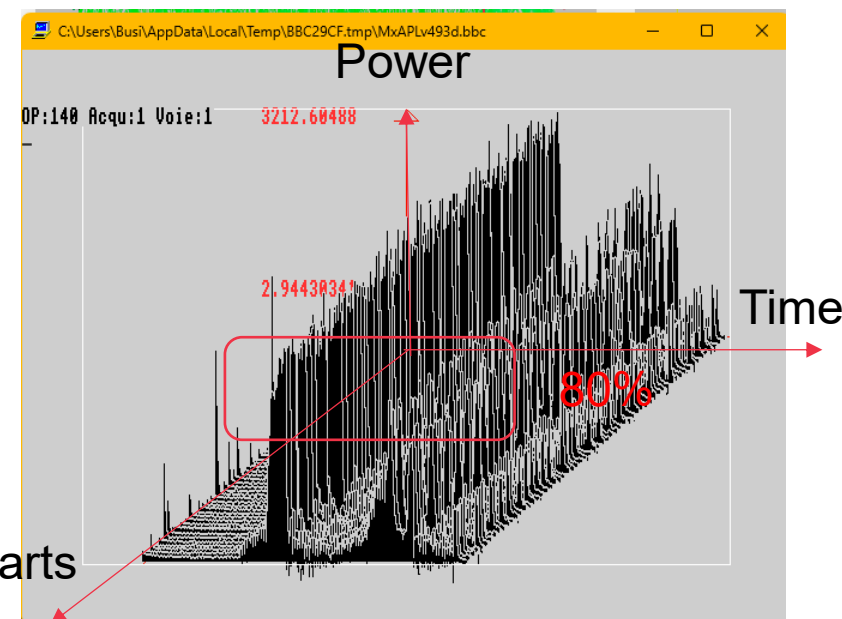
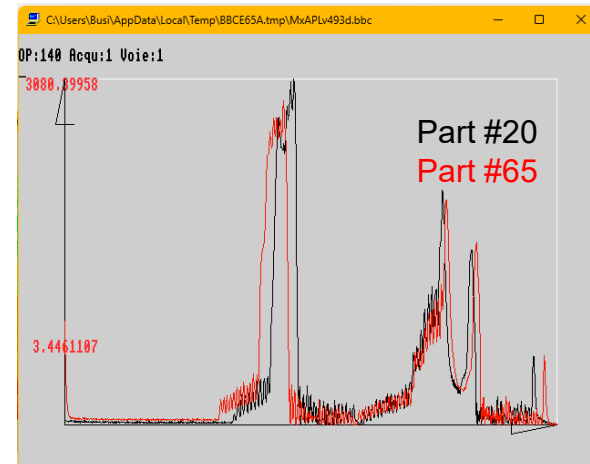
Power



Normal

80%

Before machining In machining

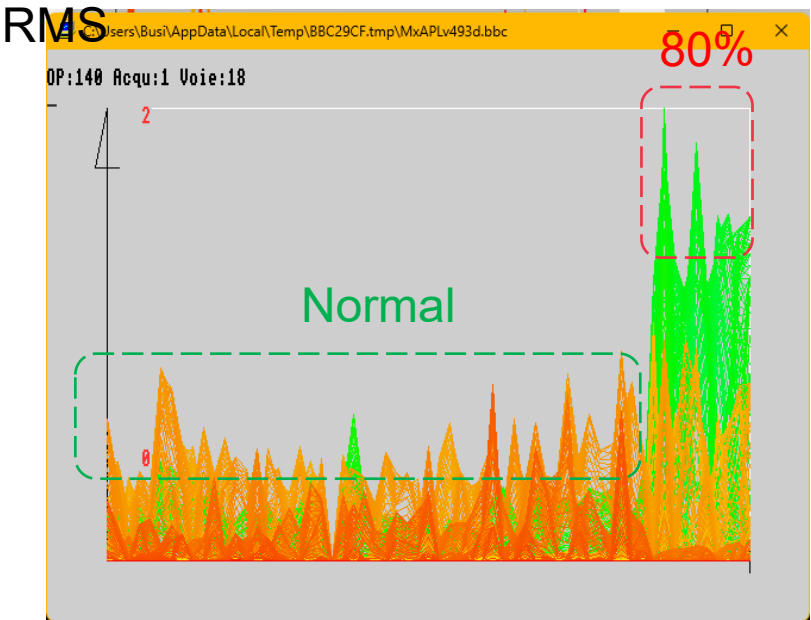


Rem : Temporal shifts sometimes !

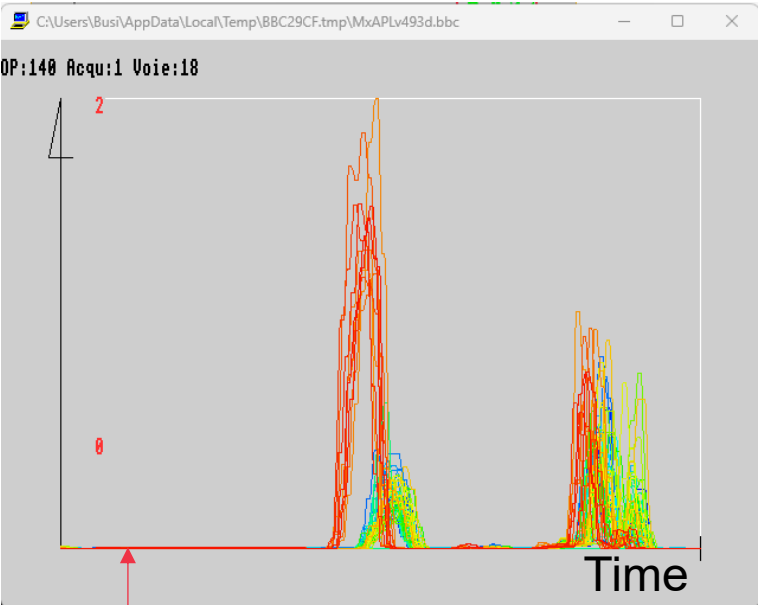
Spindle power decreases when feedrate decreases => OK

Audio : Vib 11

RMS

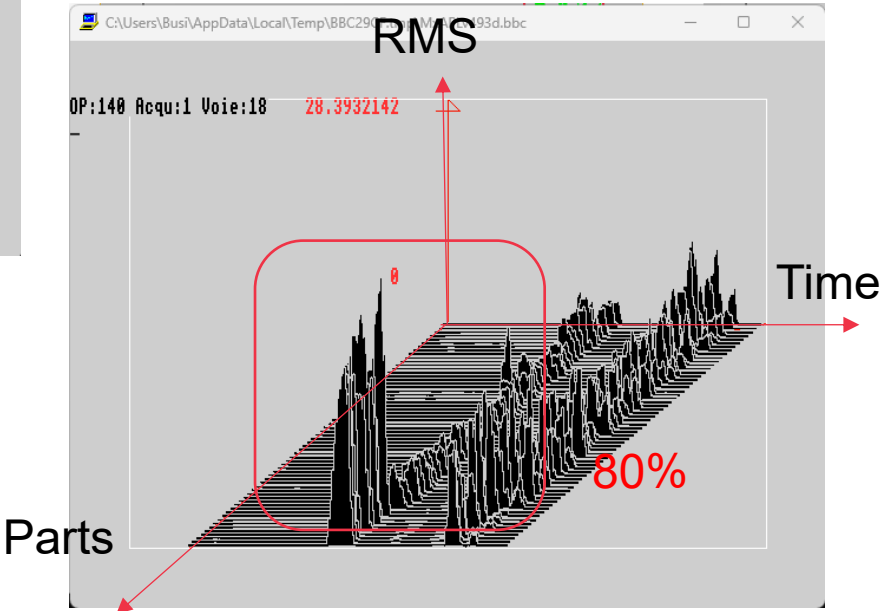
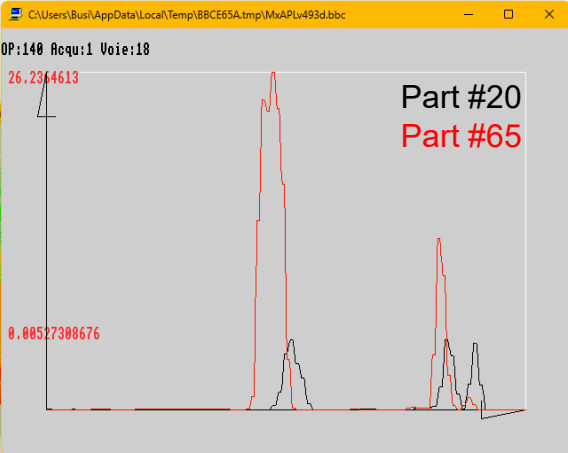


Parts



Before machining

In machining



Audio RMS, increases when feedrate decreases

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

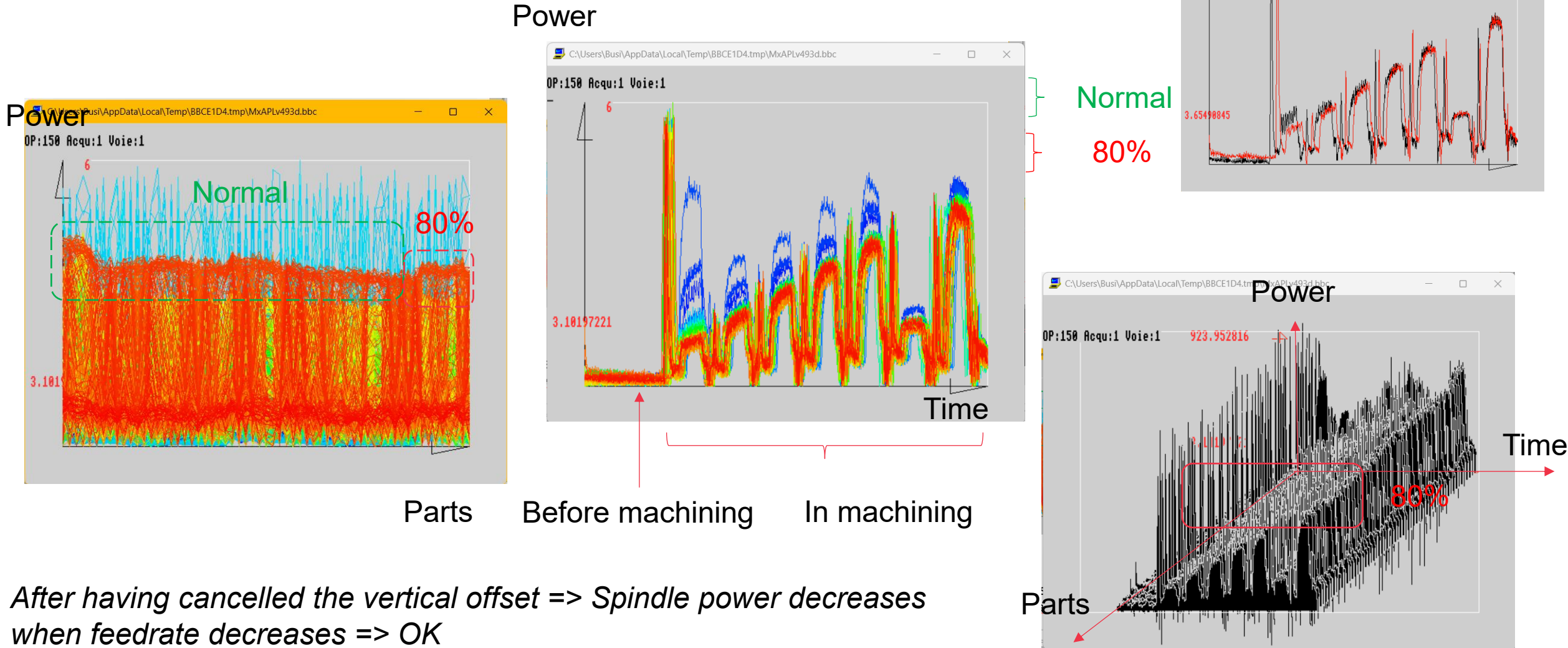
#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.099999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 150

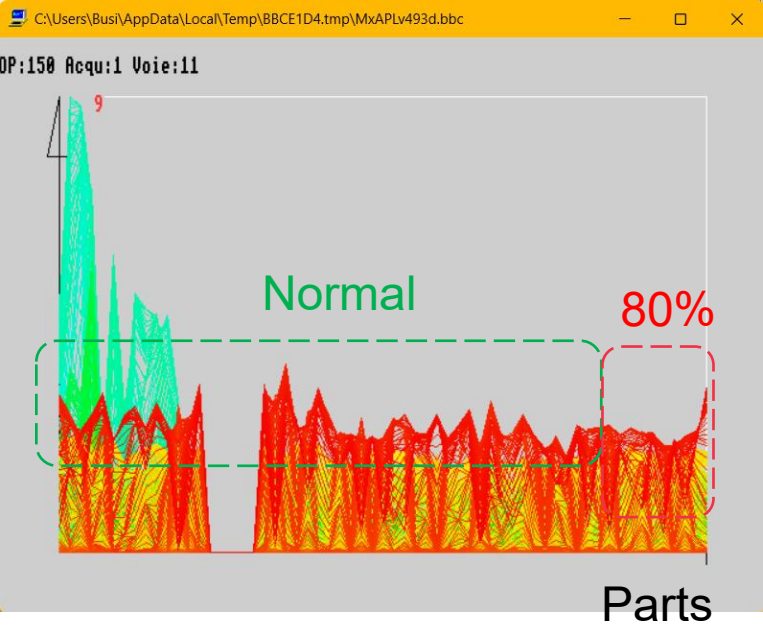
Spindle Power : Ai1



*After having cancelled the vertical offset => Spindle power decreases
when feedrate decreases => OK*

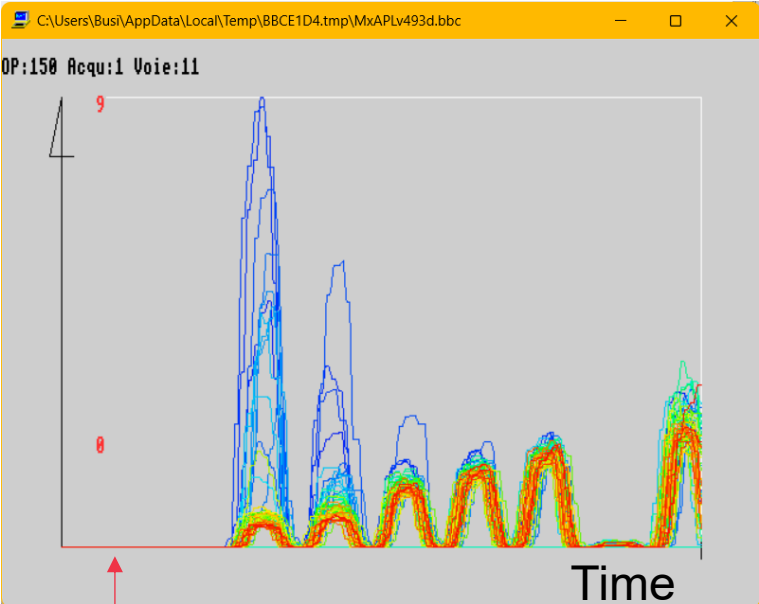
Vib : Vib 4

RMS



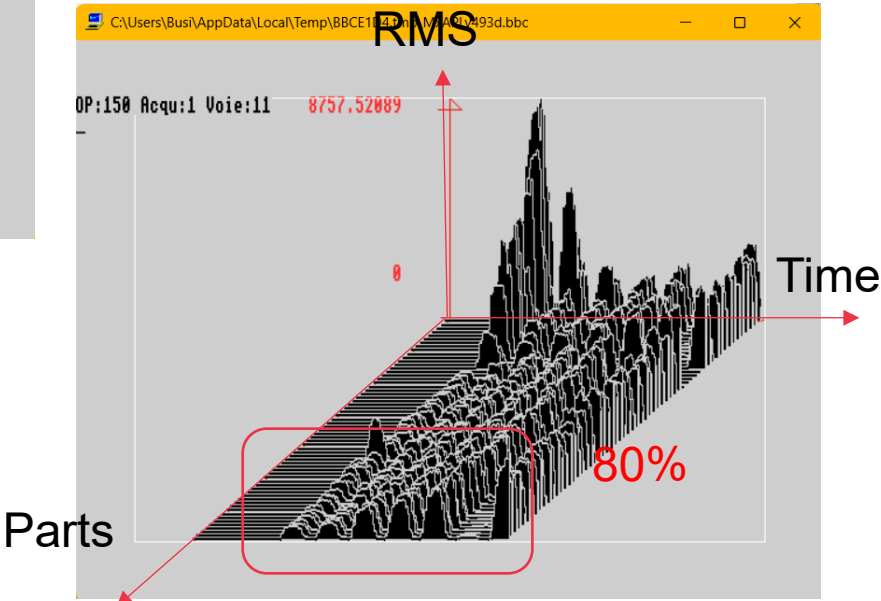
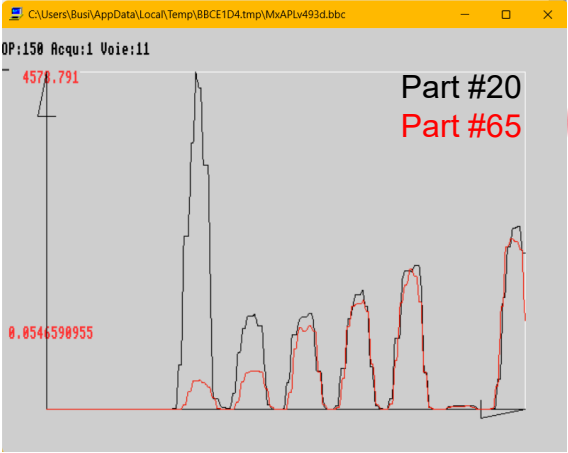
RMS, decreases a little when feedrate decreases

RMS



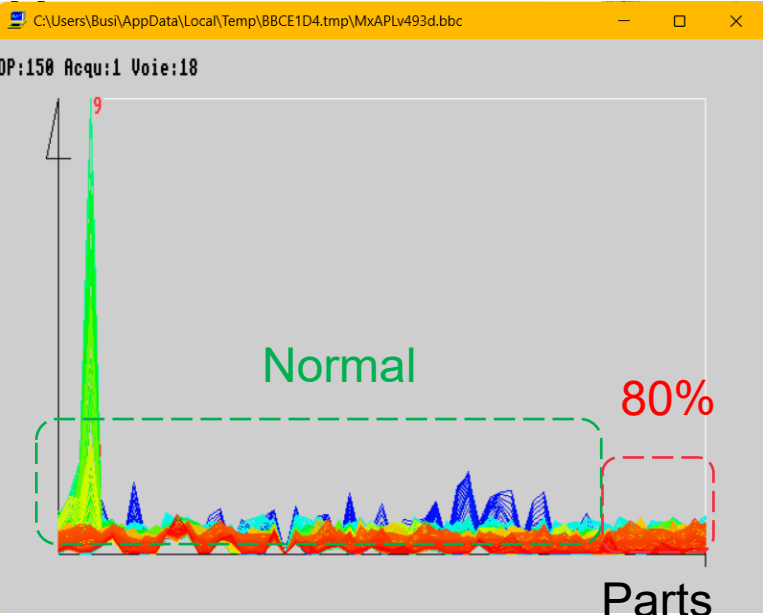
Before machining

In machining



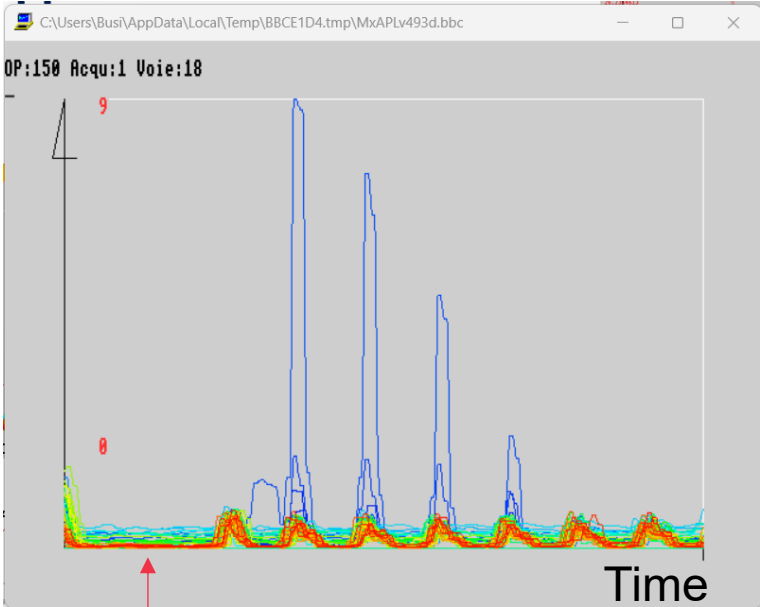
Audio : Vib 11

RMS



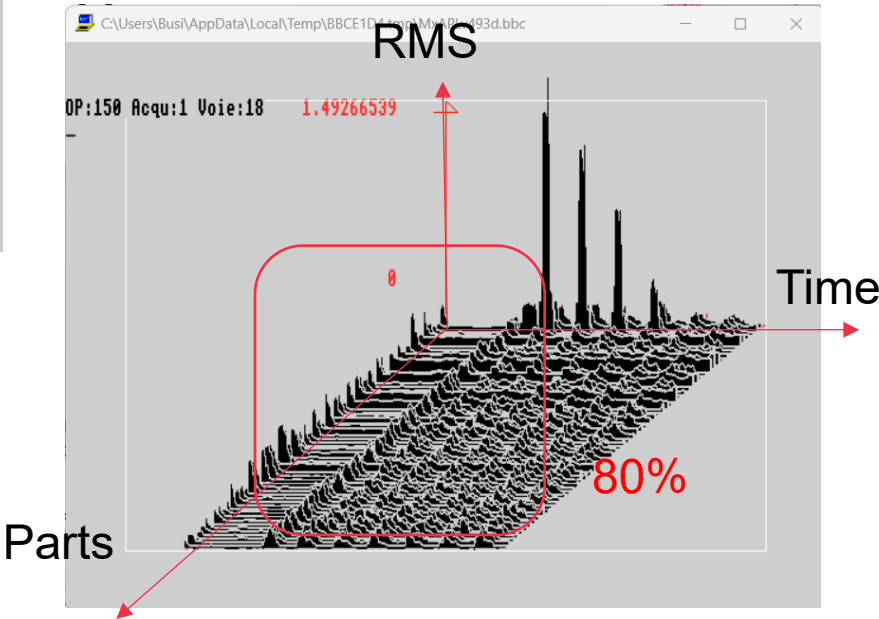
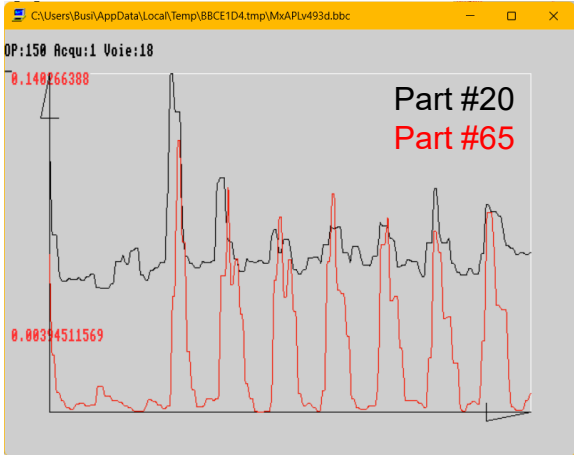
Audio RMS, decreases when feedrate decreases

RMS

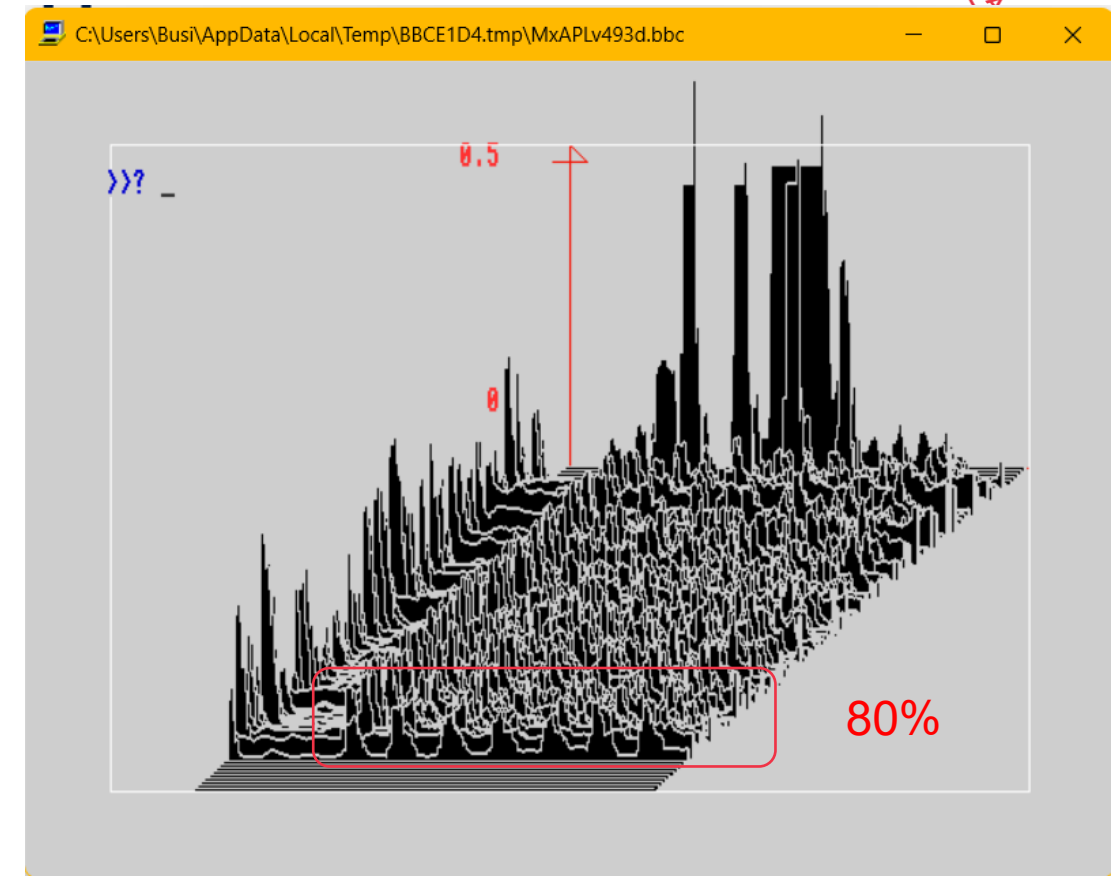
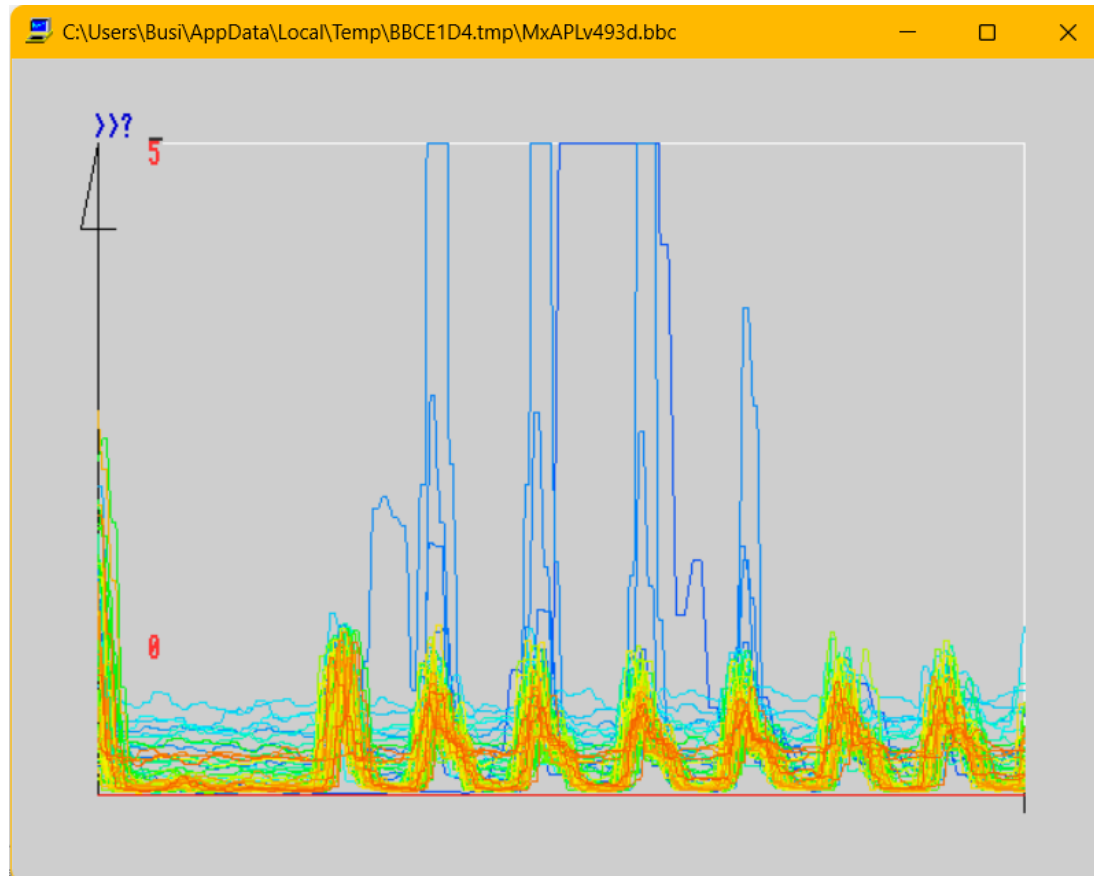


Before machining

In machining



Audio : Vib 11



Even with a vertical zoom => Difficult to see here differences when en feedrate decreases (RMS lower ?)

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

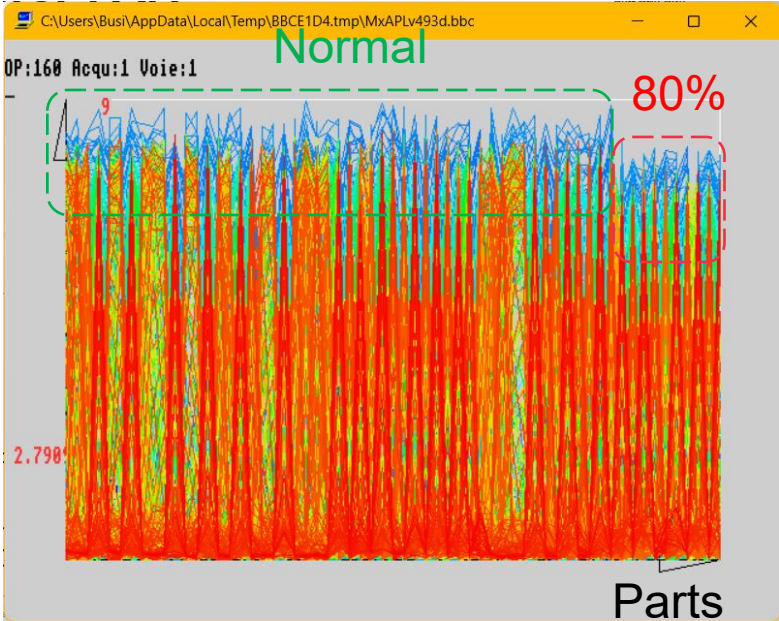
```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.099999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" /> Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

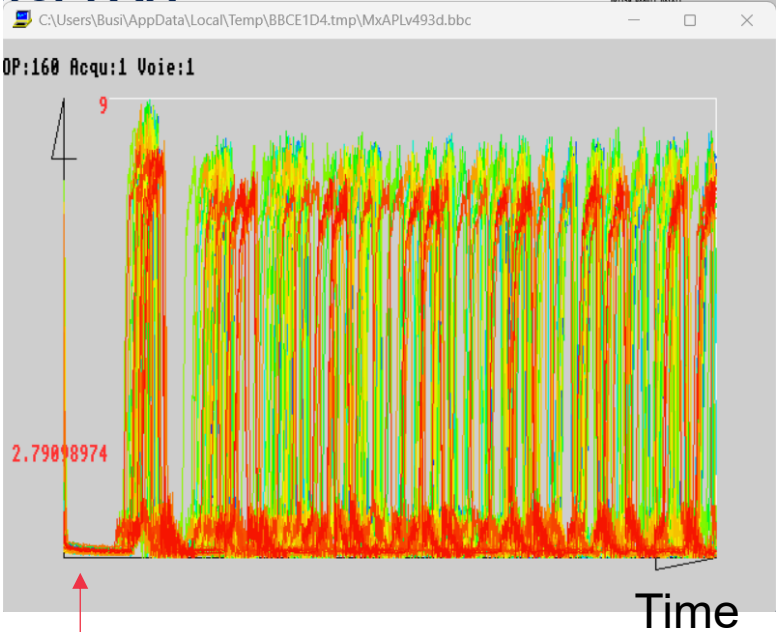
Machining OP 160

Spindle Power : Ai1

Power



Power

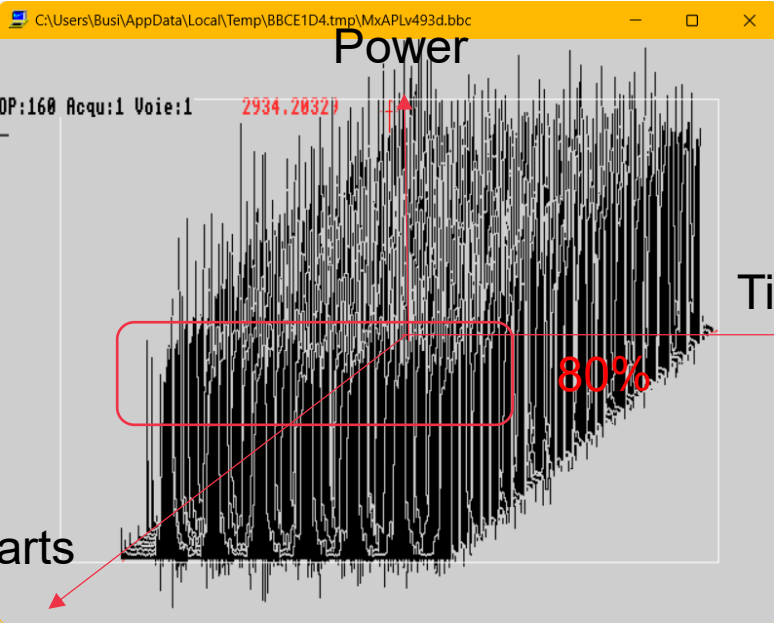
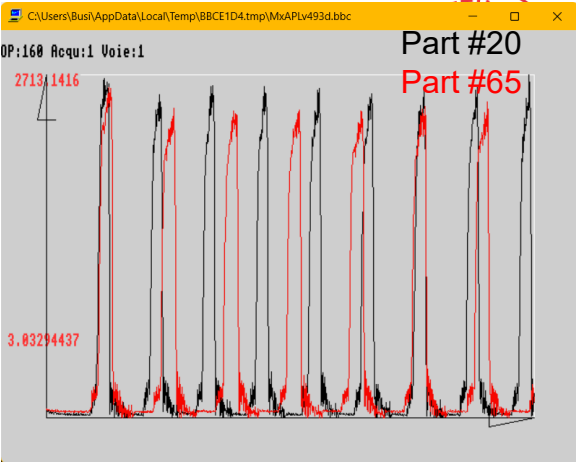


Before machining

In machining

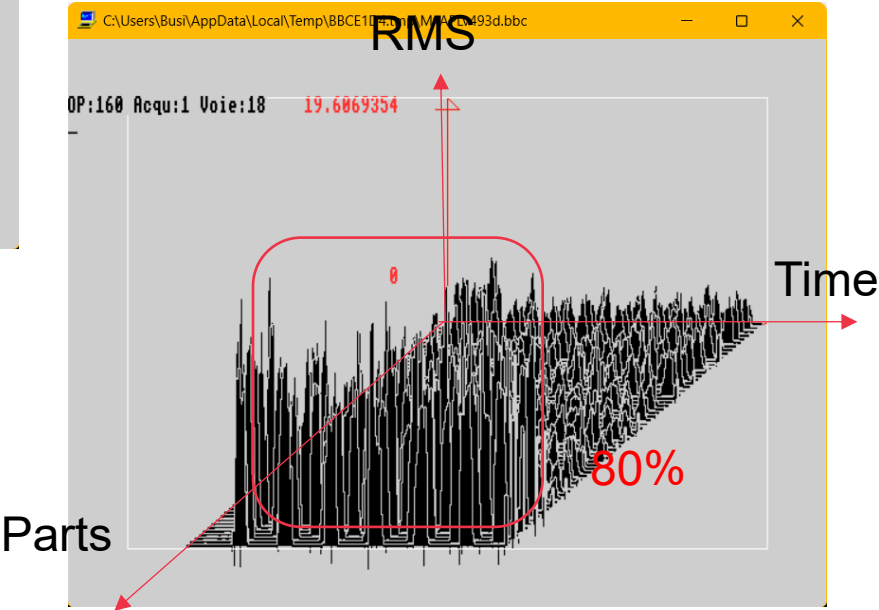
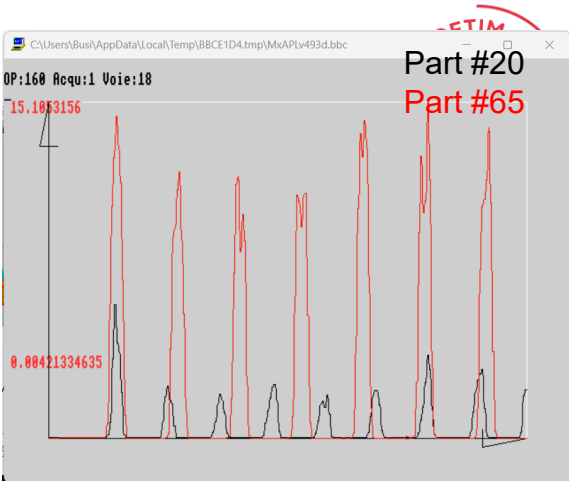
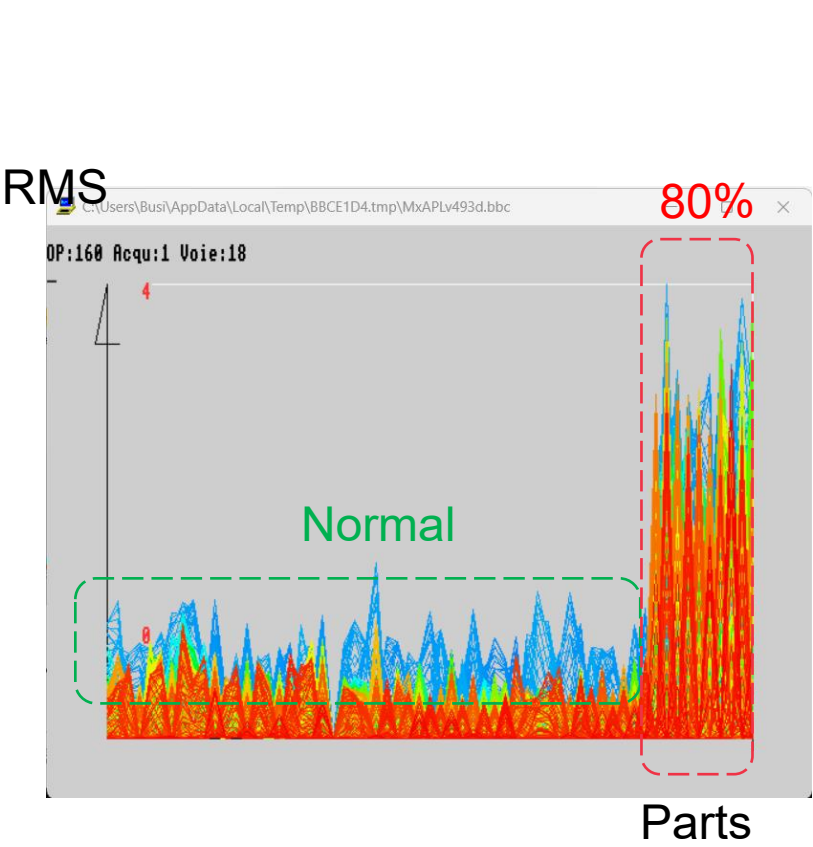
Normal

80%



Spindle power decreases when feedrate decreases => OK

Audio : Vib 11



Audio RMS, increases when feedrate decreases

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

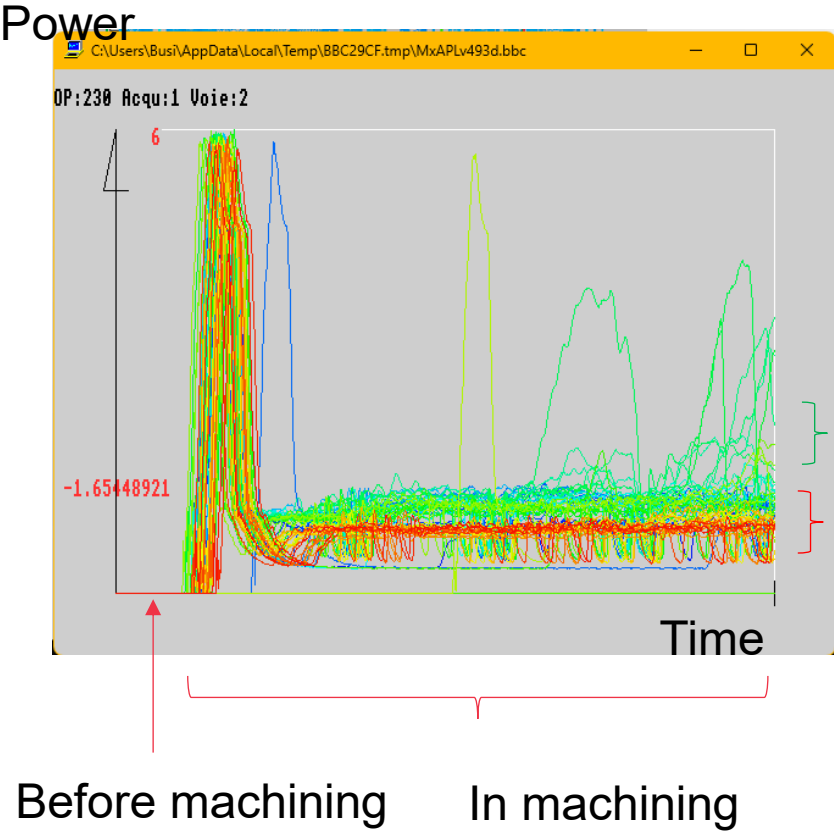
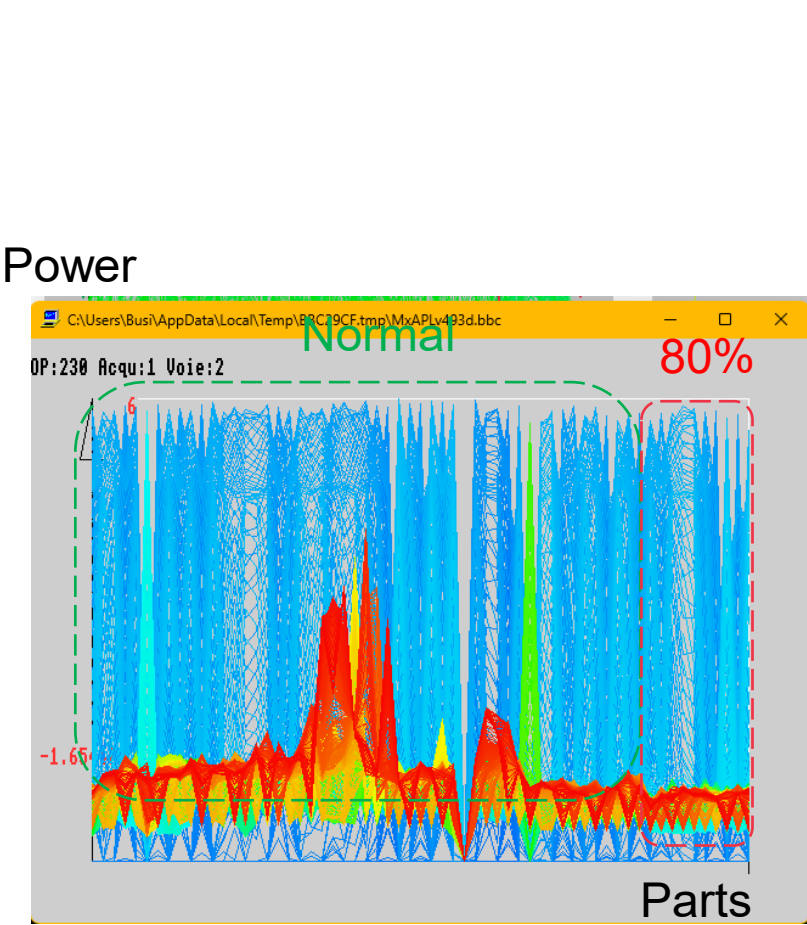
#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

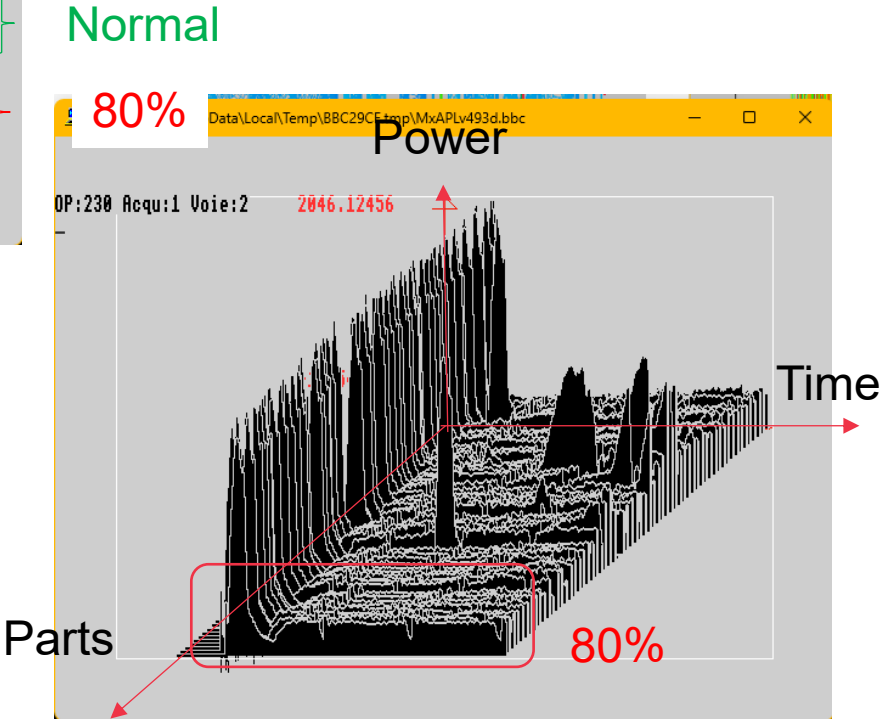
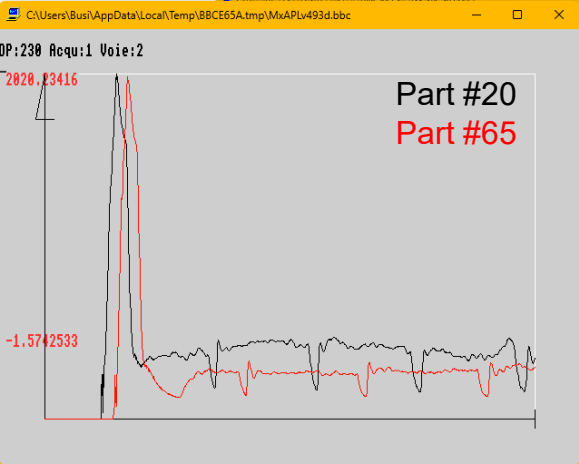
```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 230

Spindle Power : Ai2



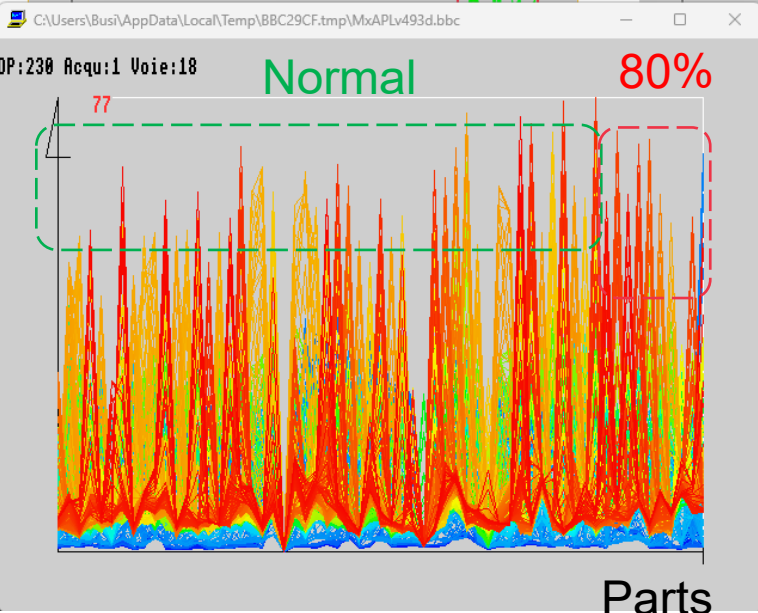
Before machining In machining



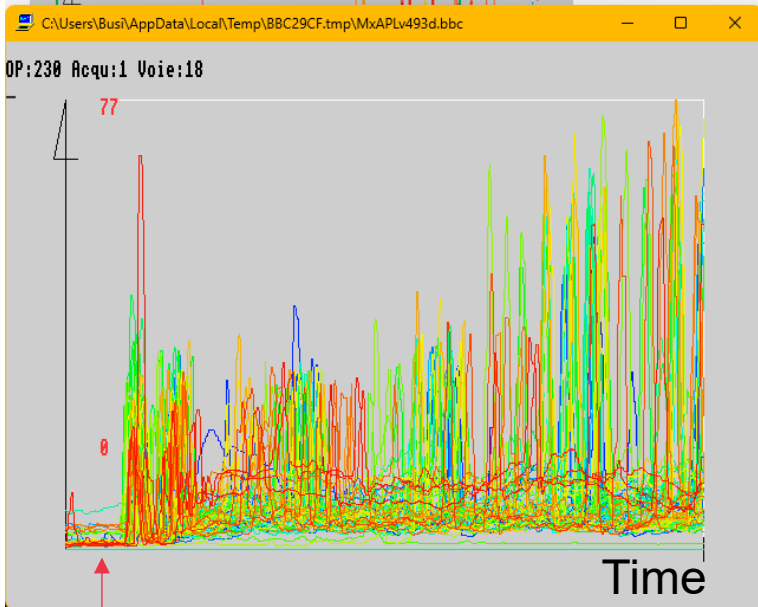
Rem : Temporal shifts sometimes !
Spindle power decreases when feedrate decreases => OK

Audio : Vib 11

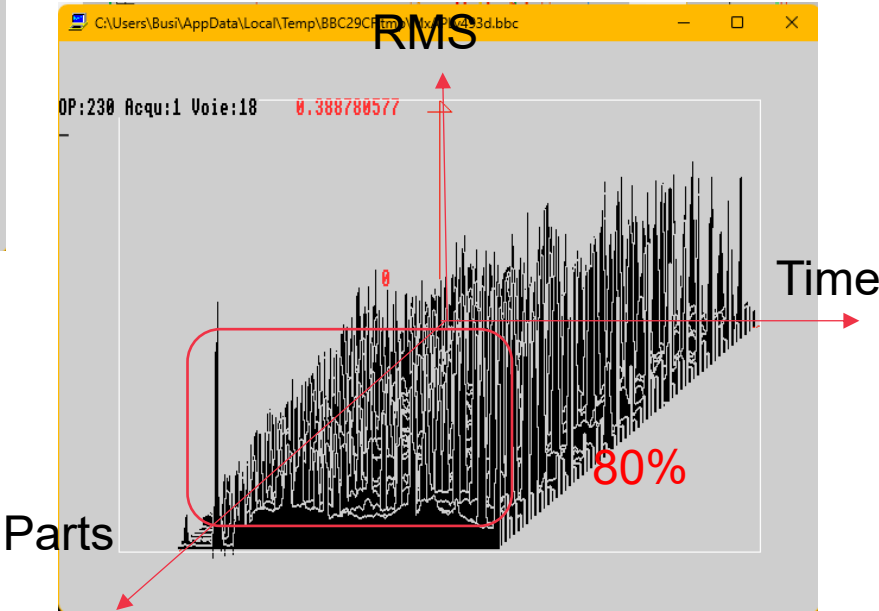
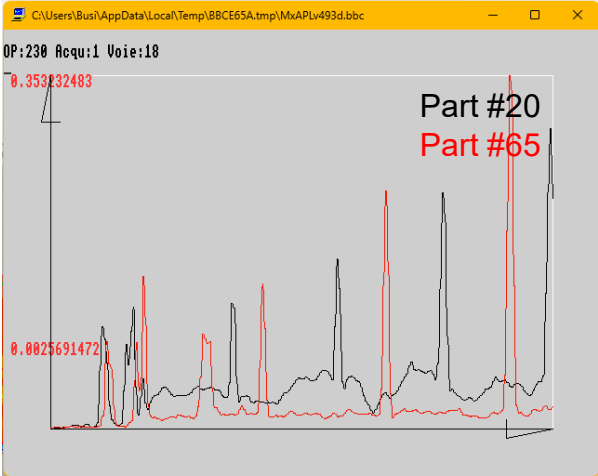
RMS



RMS



Before machining In machining



Audio RMS, remains the same when feedrate decreases ...

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

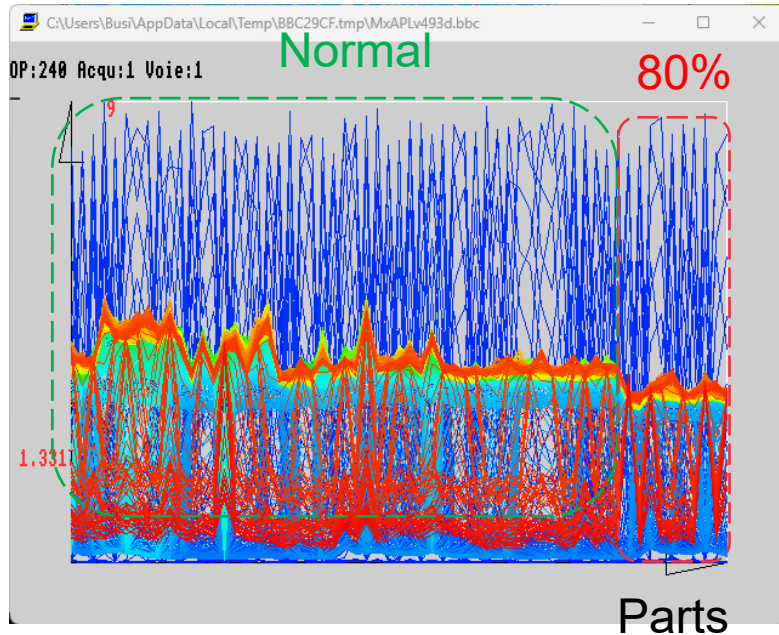
```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
Le 26/01/26
Intermediate report Project n°316285 - Date 26/01/26 Page 91/188
```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
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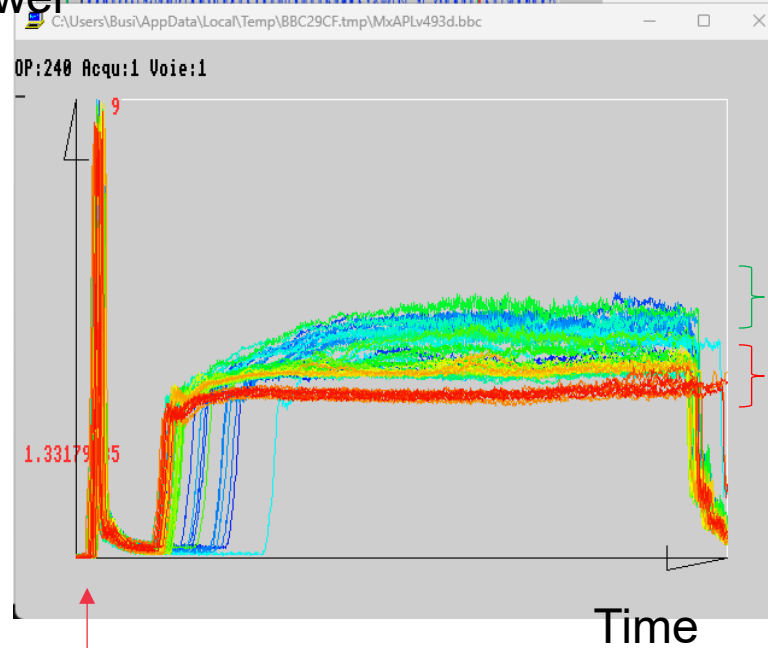
Machining OP 240

Spindle Power : Ai1

Power

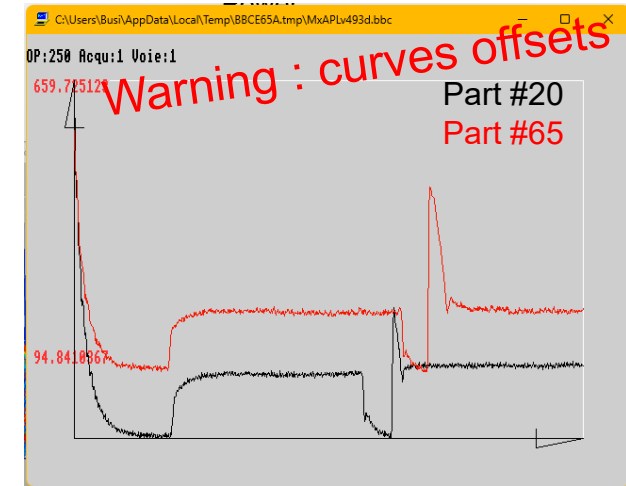


Power



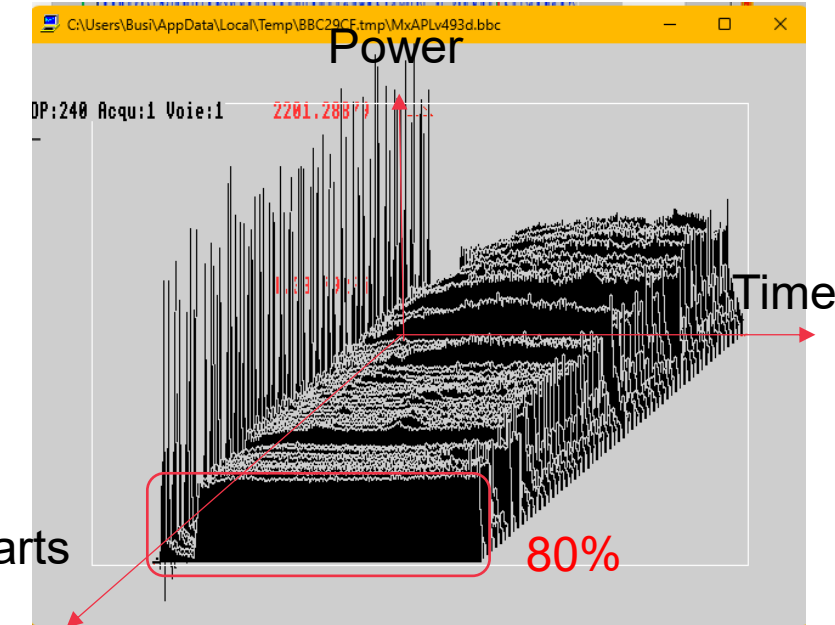
Before machining

In machining



Normal

80%



Parts

80%

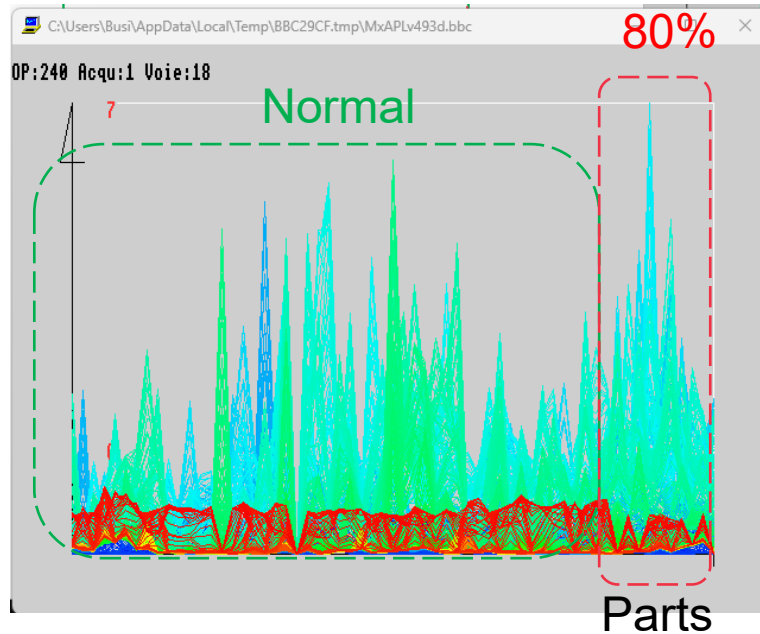
Rem : Temporal shifts sometimes !

Cutting time increases when feedrate decreases => OK

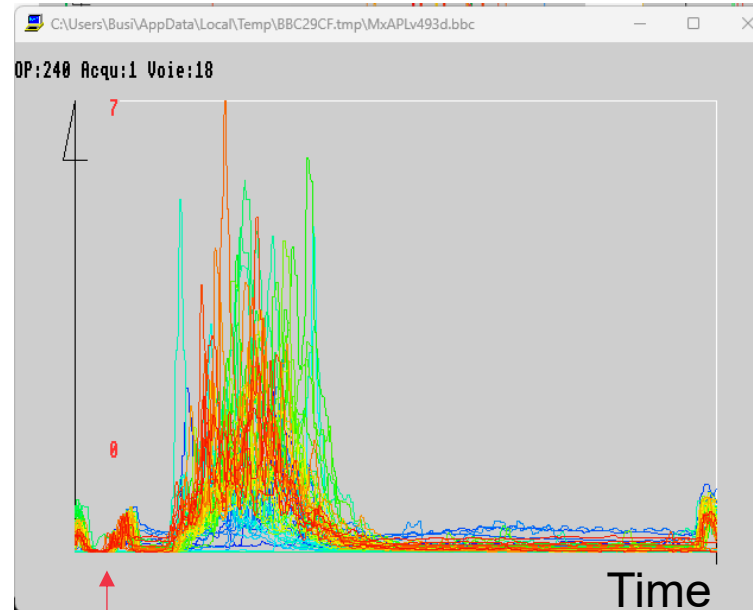
Spindle power decreases when feedrate decreases => OK

Audio : Vib 11

RMS



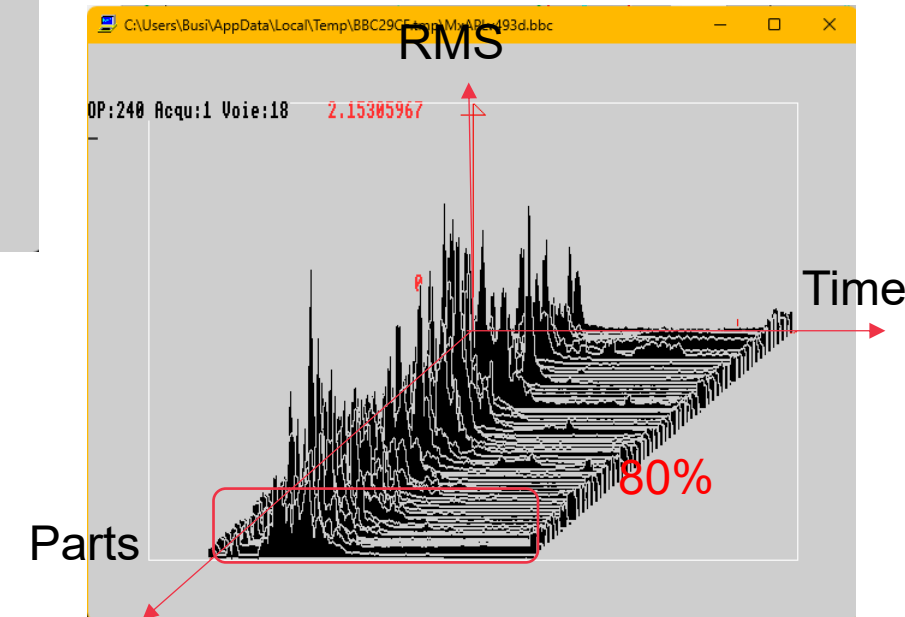
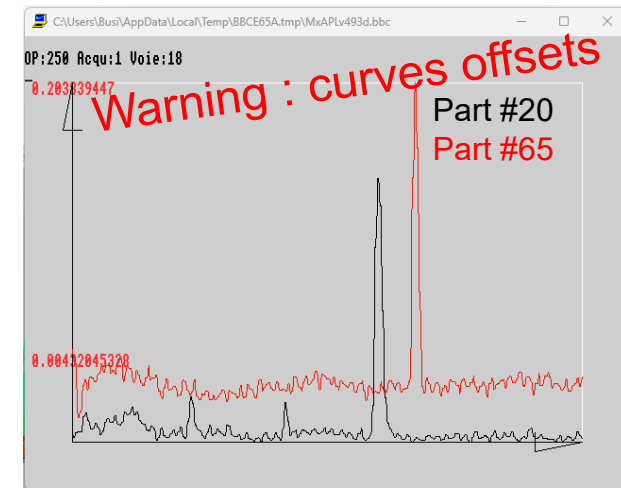
RMS



Before machining

In machining

Audio RMS, seems to decrease when feedrate decreases for some times slices... => Not obvious



Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

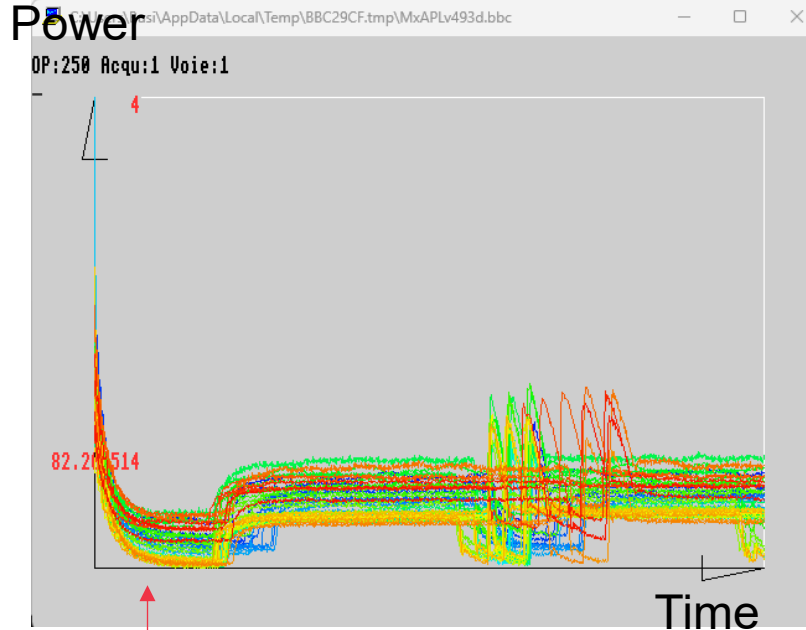
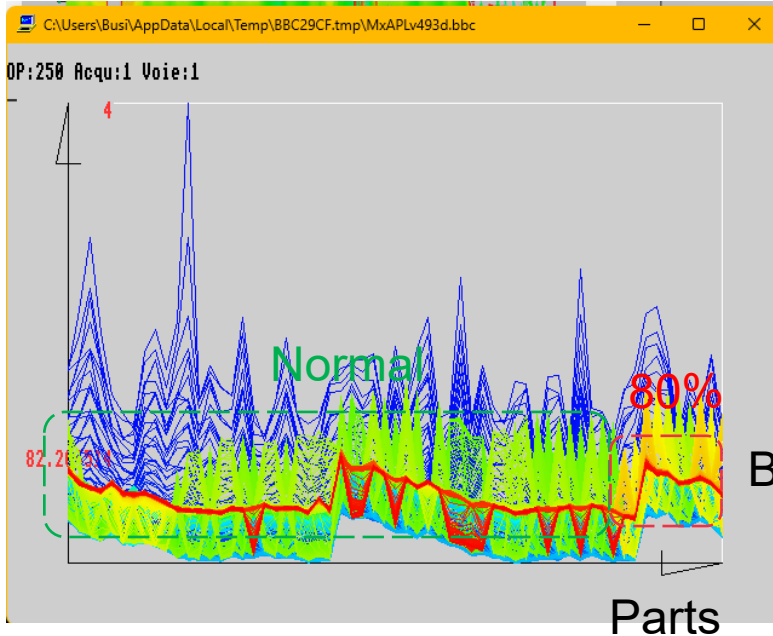
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Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 250

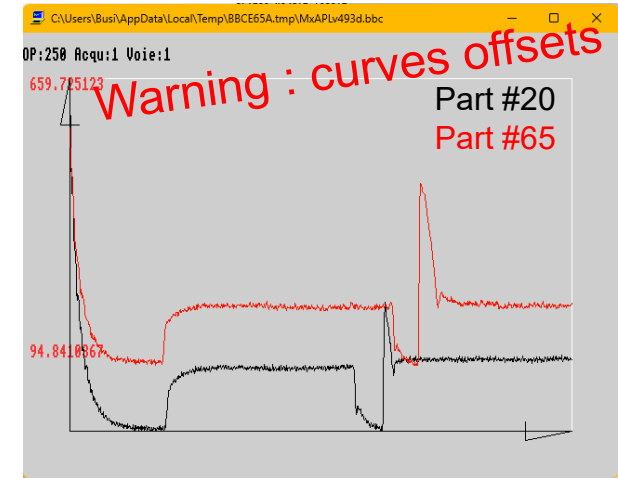
Spindle Power : Ai1

Power

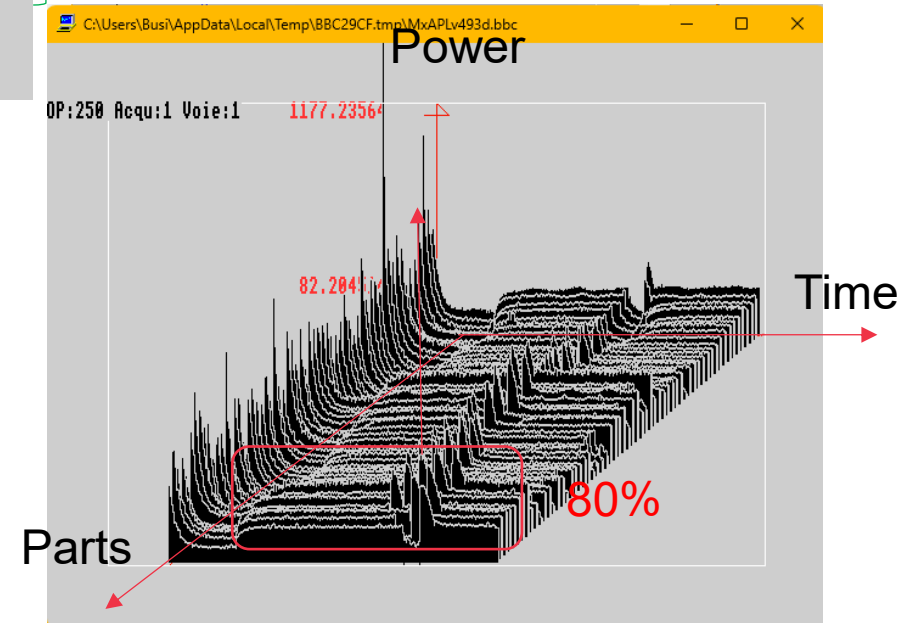


Before machining

In machining



80%
Normal



*Rem : Temporal shifts sometimes !
Spindle power seems to be remains the same or decreases a little when
feedrate decreases*

Vib : Vib3



Artefact => To Do again

Audio : Vib 11



Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
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Duration="6.0999999999999979" TemplateComputed="true">
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<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
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<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" /> Le 26/01/26
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```

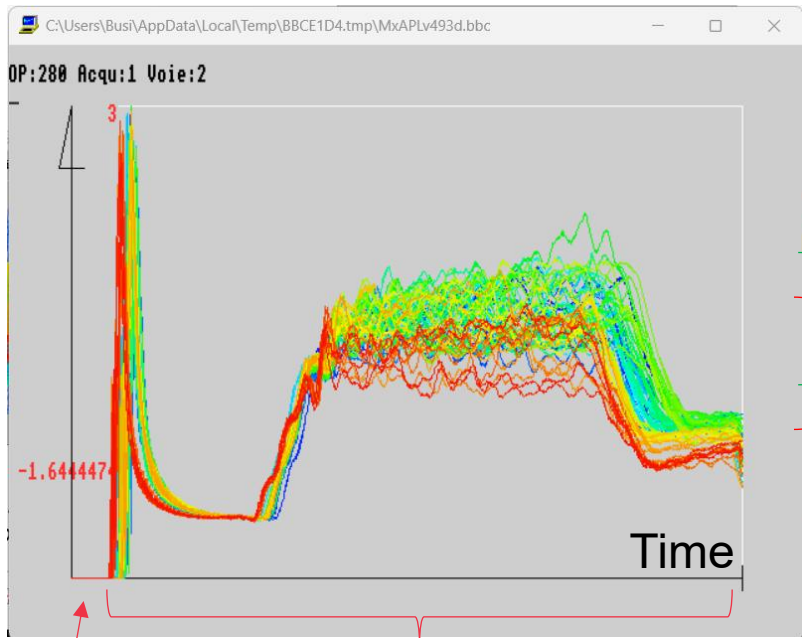
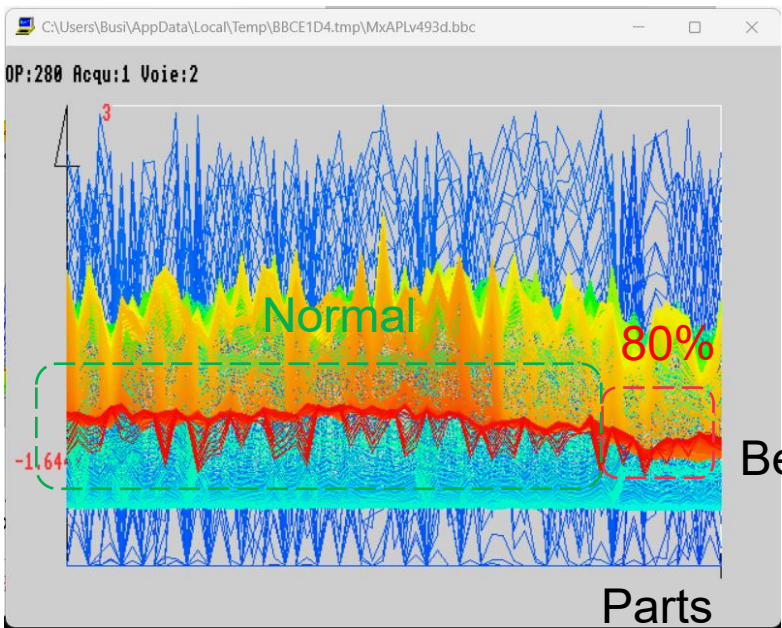
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<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 280

Spindle Power : Ai2

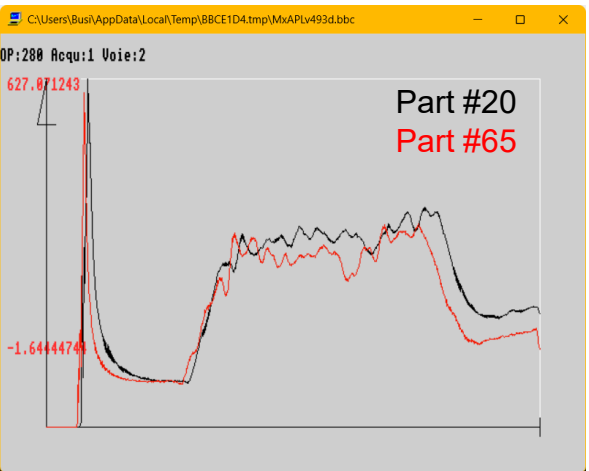
Power

Power



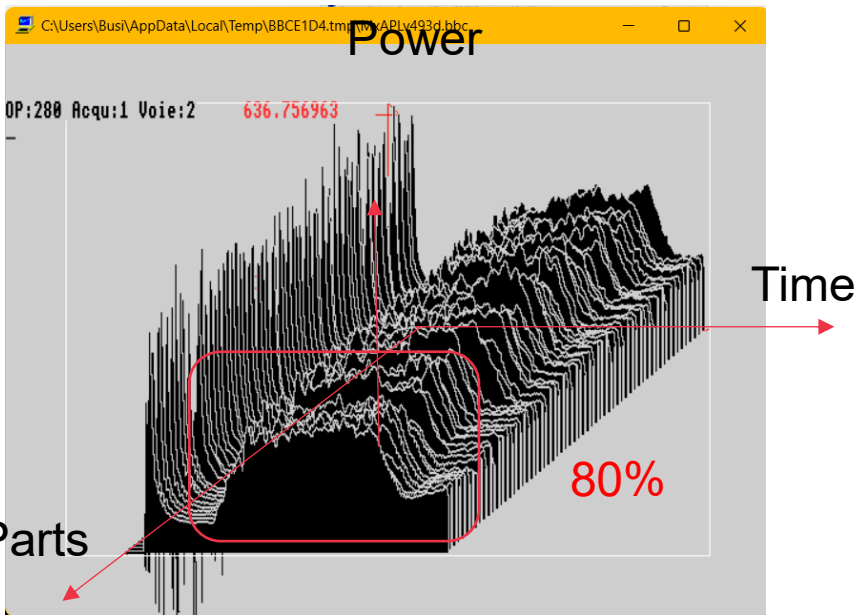
Before machining

In machining



Normal

80%



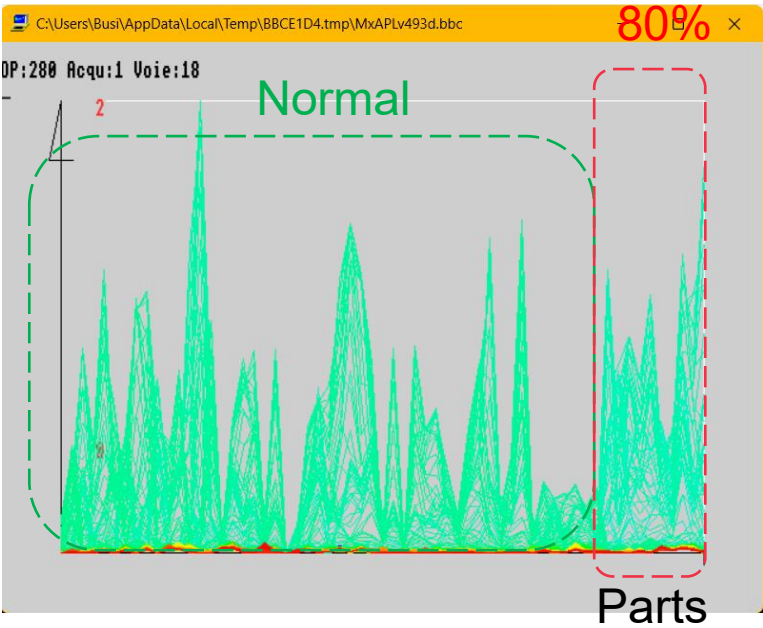
Parts

Time

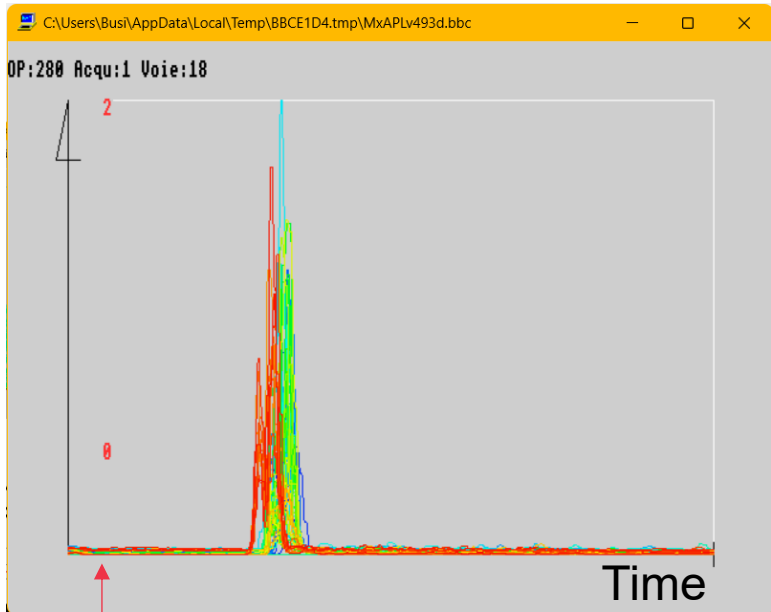
Spindle power decreases when feedrate decreases => OK

Audio : Vib 11

RMS

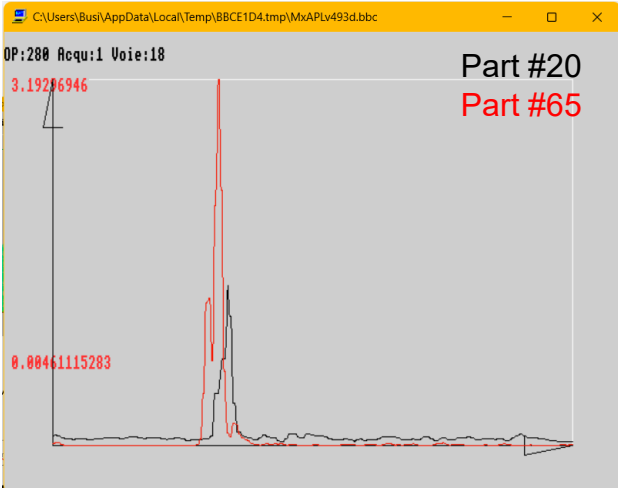


RMS

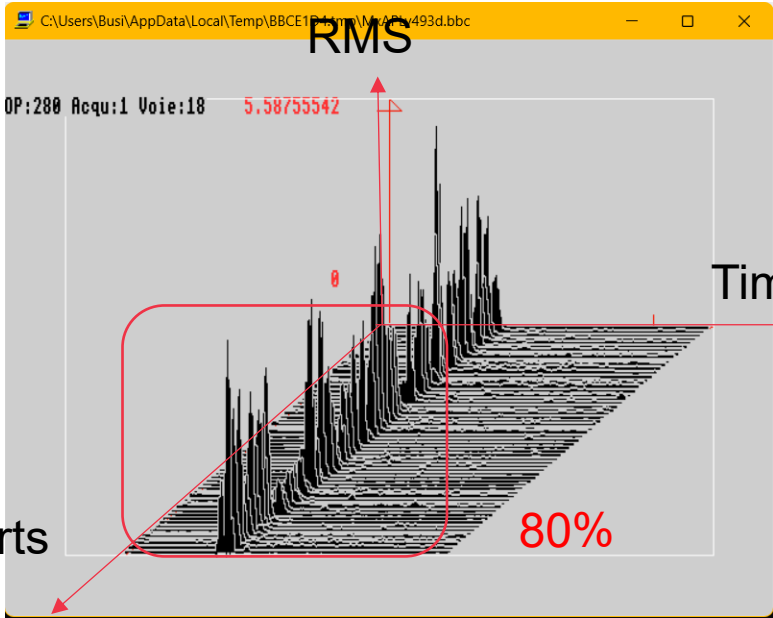


Before machining

In machining



RMS



Audio RMS, seems to increase when feedrate decreases ?

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

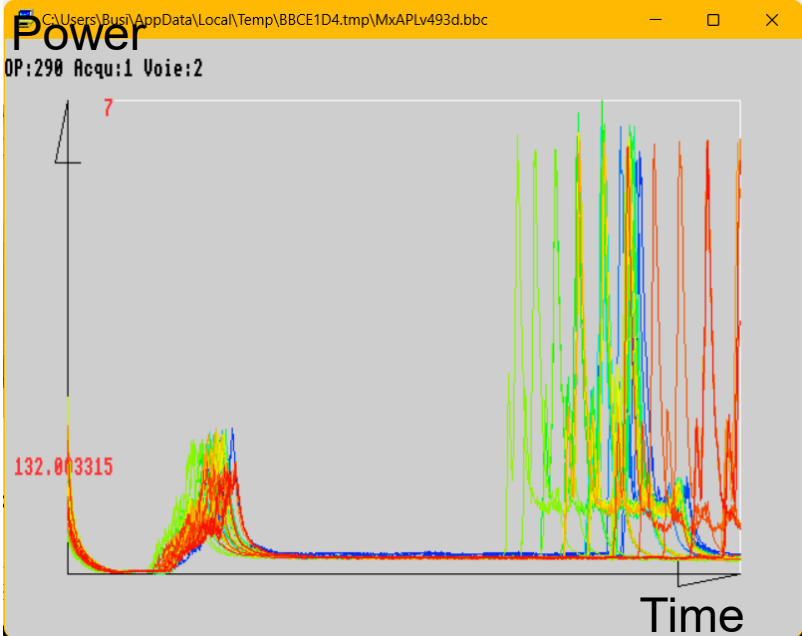
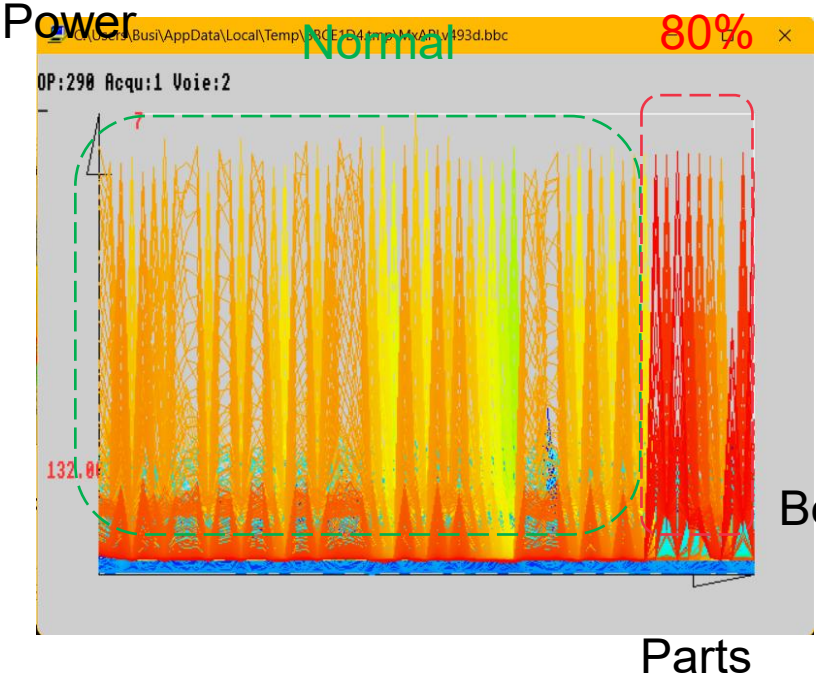
#18 : Vib11 (audio micro)

```
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<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.099999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
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<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
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<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
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```

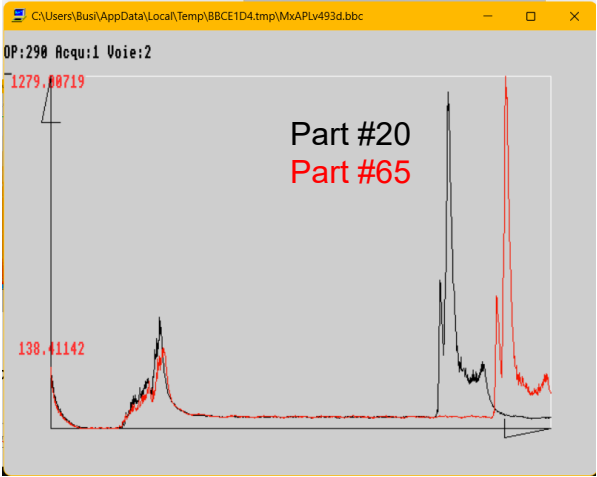
Machining OP 290

Spindle Power : Ai2

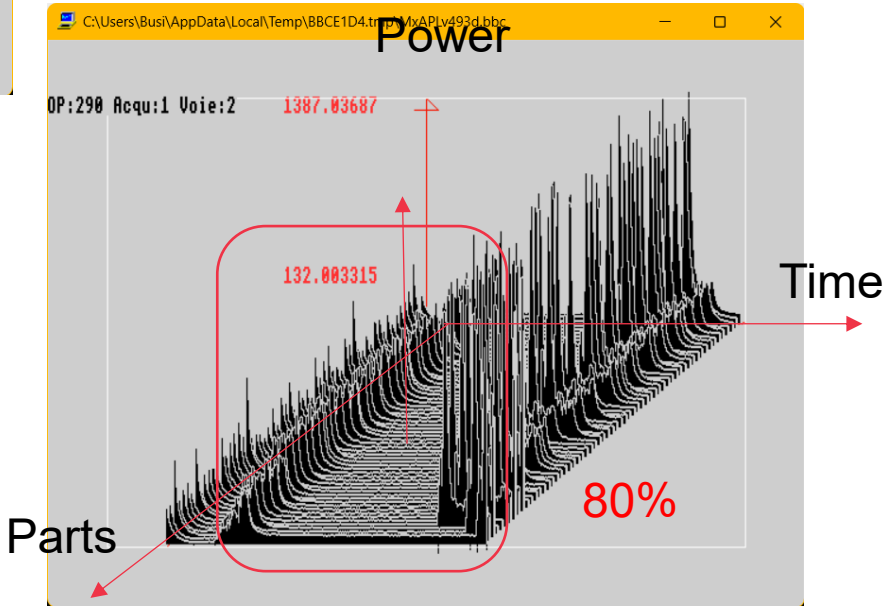


Before machining

In machining



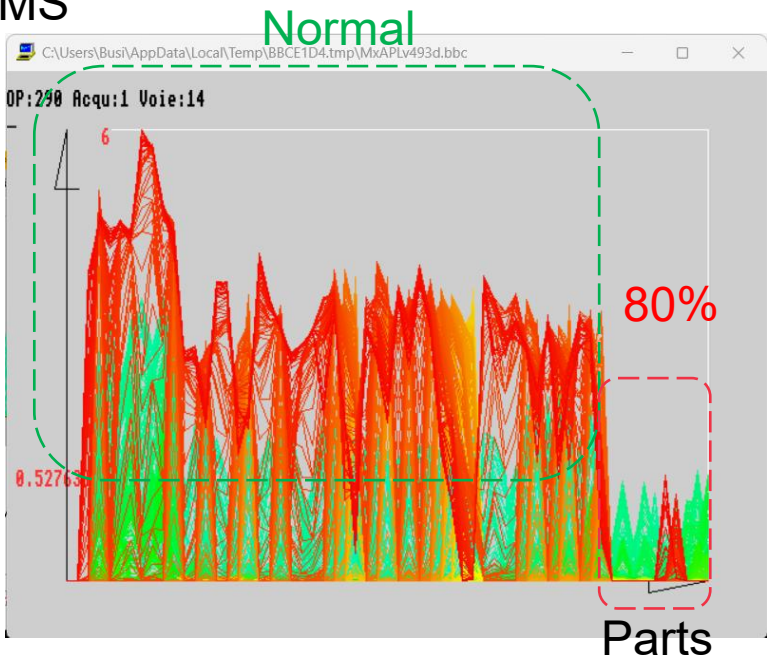
Normal
80%



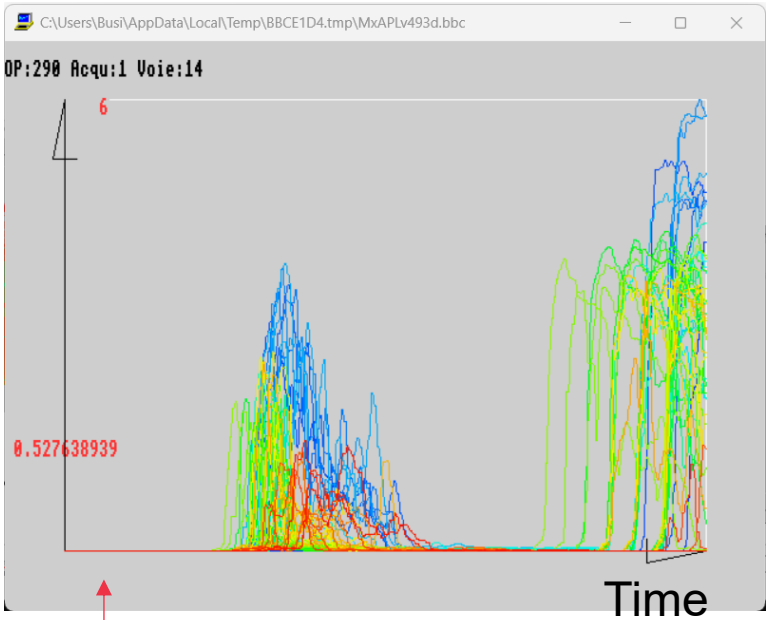
Spindle power increases a little when feedrate decreases ?

Vib : Vib 7

RMS



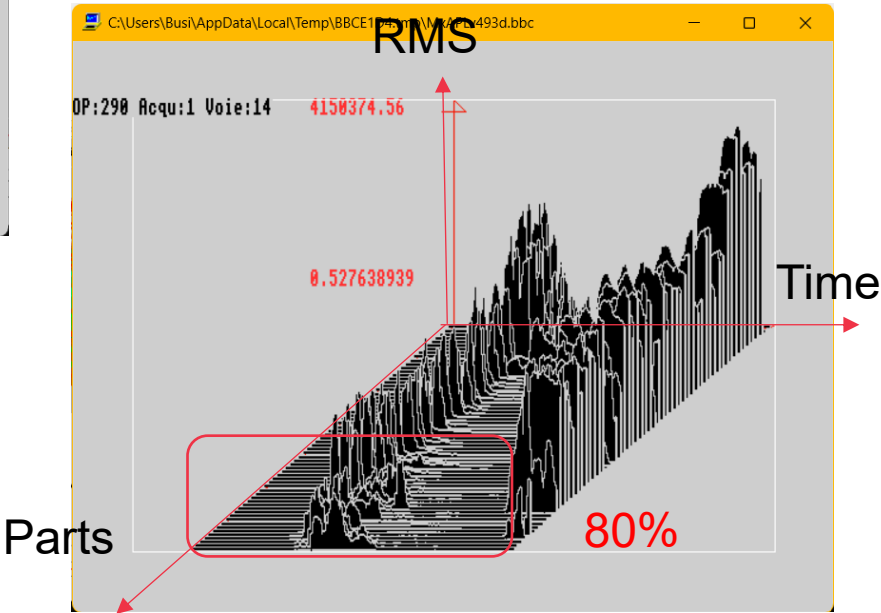
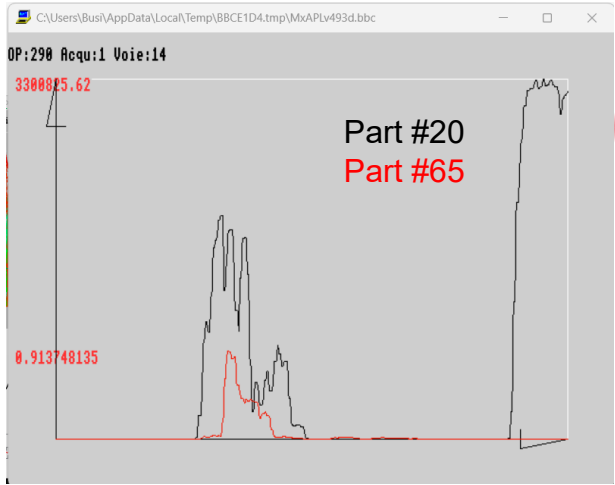
RMS



Before machining

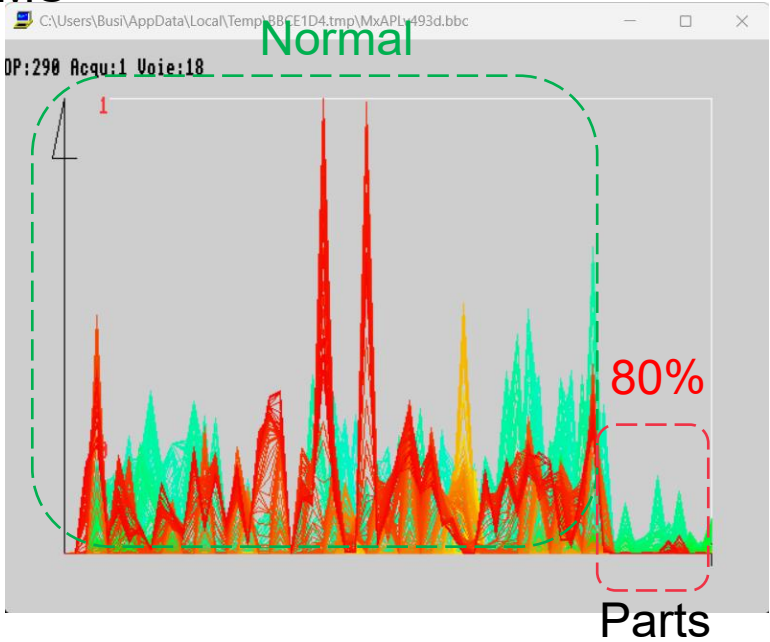
In machining

RMS decreases when feedrate decreases

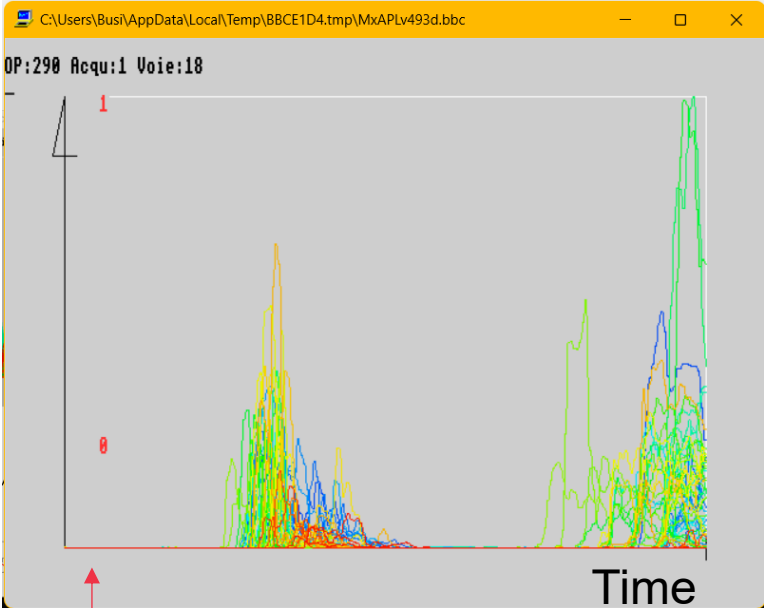


Audio : Vib 11

RMS

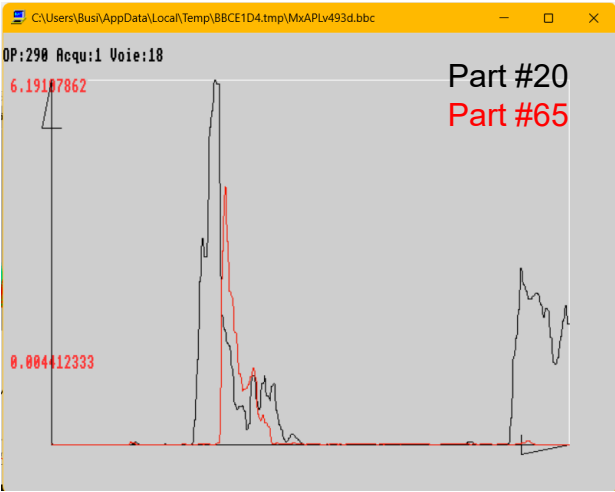


RMS



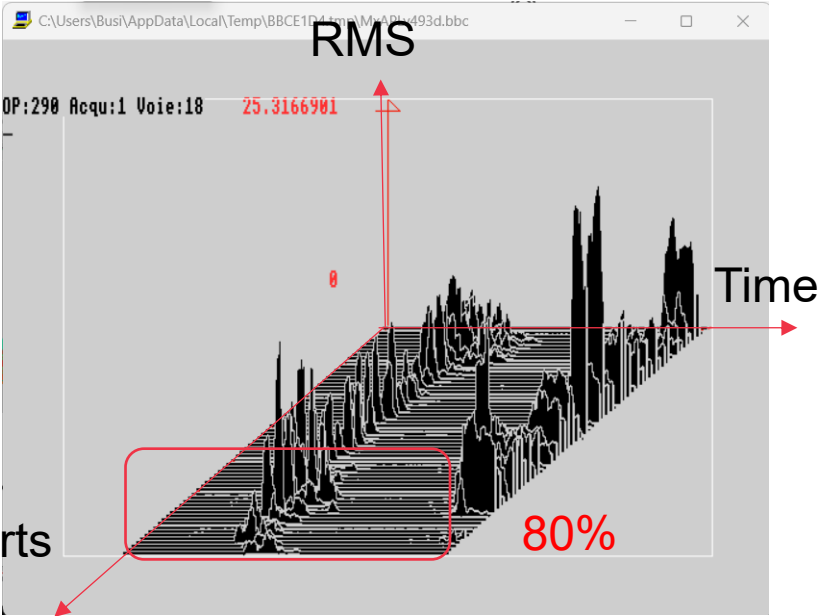
Before machining

In machining



Part #20
Part #65

RMS



Audio RMS, seems to increase when feedrate decreases ?

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
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<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
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<Operation id="240" description="PERCAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
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<Operation id="260" description="AL F" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="PERCAGE DIAM 10" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 311

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" /> Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

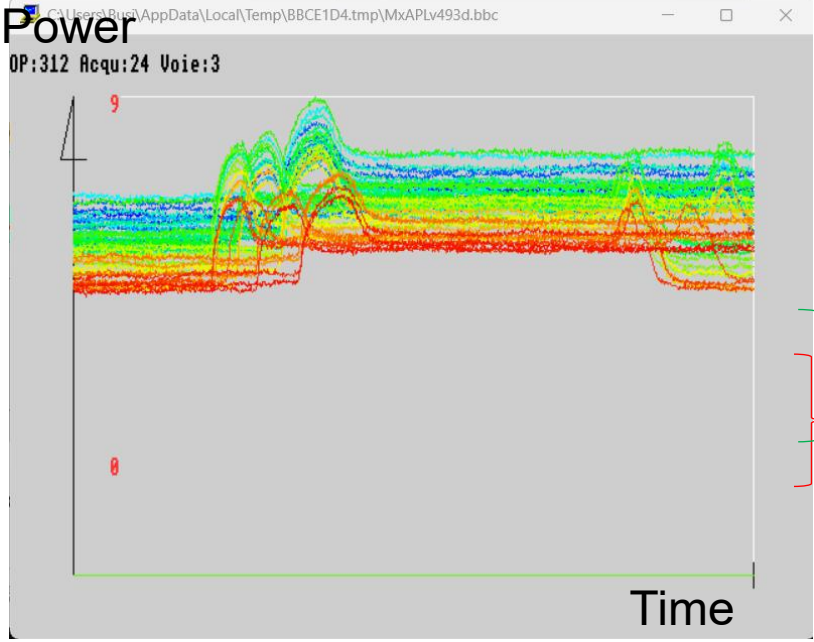
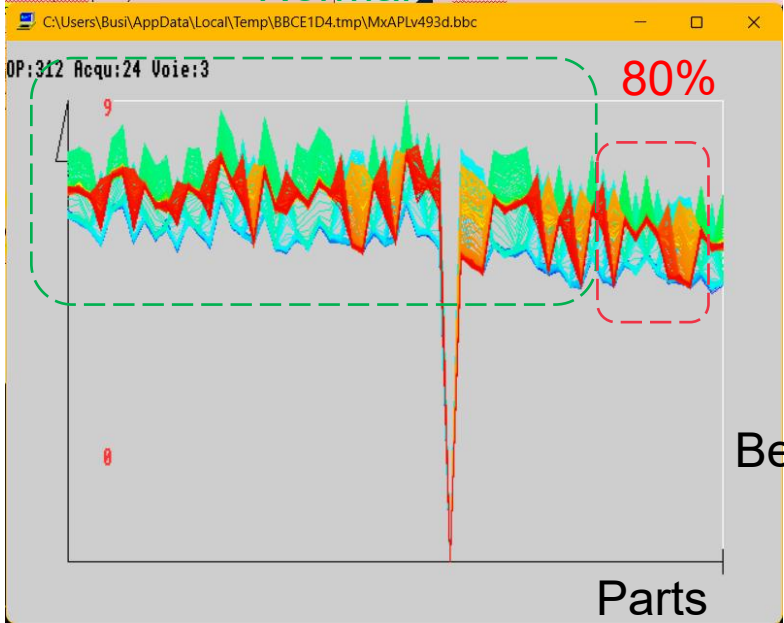
Machining OP 312

(last pass #24)

Spindle Power : Ai3

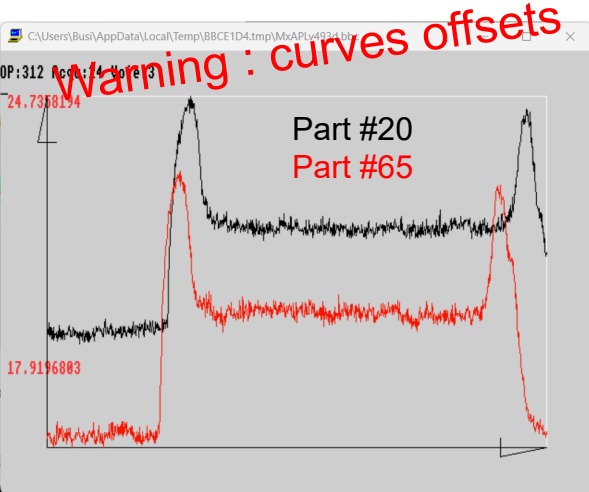
Power

Normal

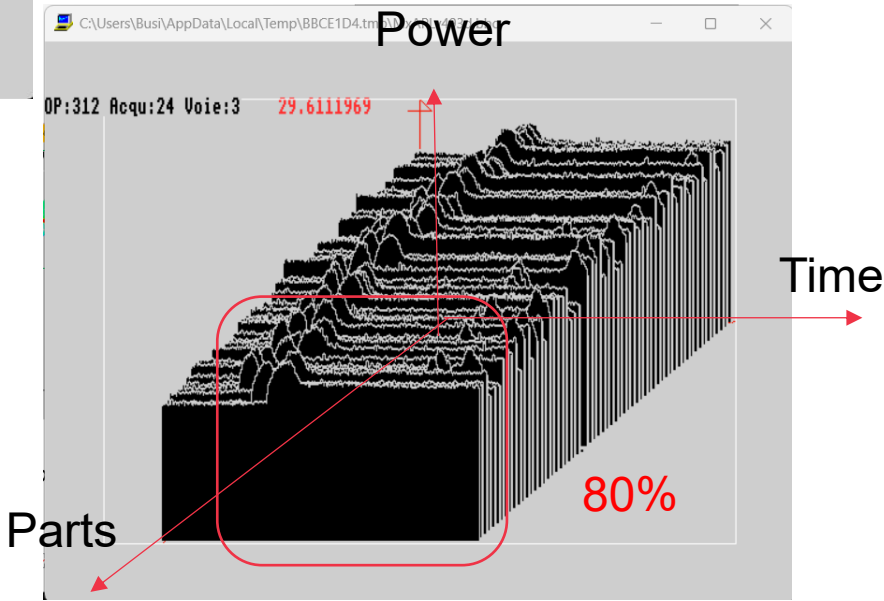


Before machining

In machining



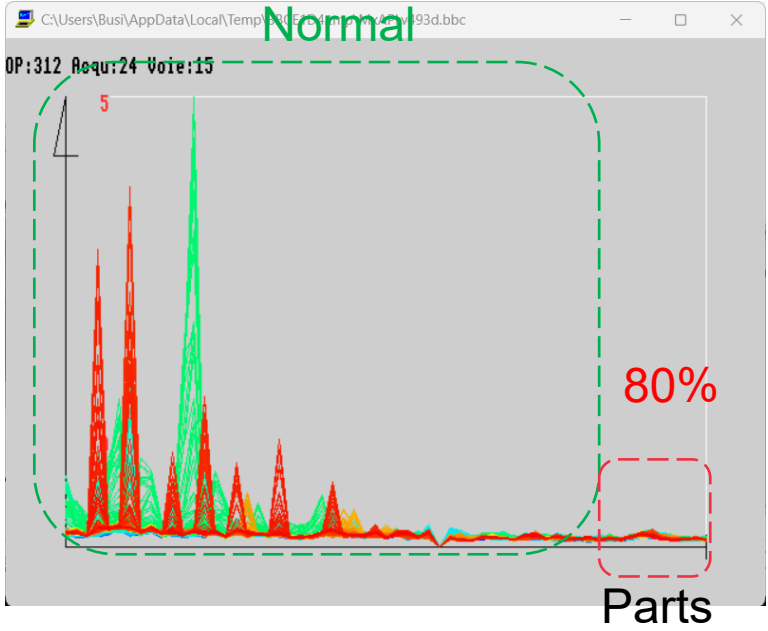
Normal
80%



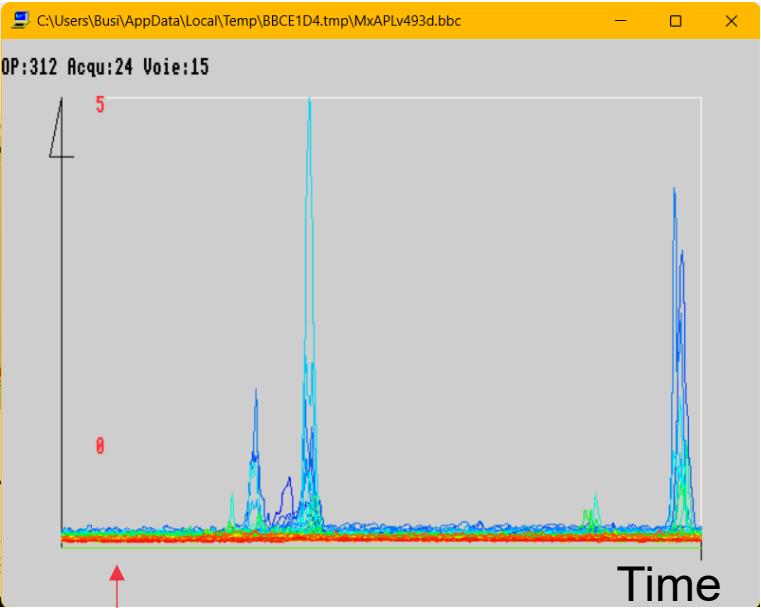
Warning to the vertical offset : Spindle power decreases a little when feedrate decreases ?

Vib : Vib 8

RMS

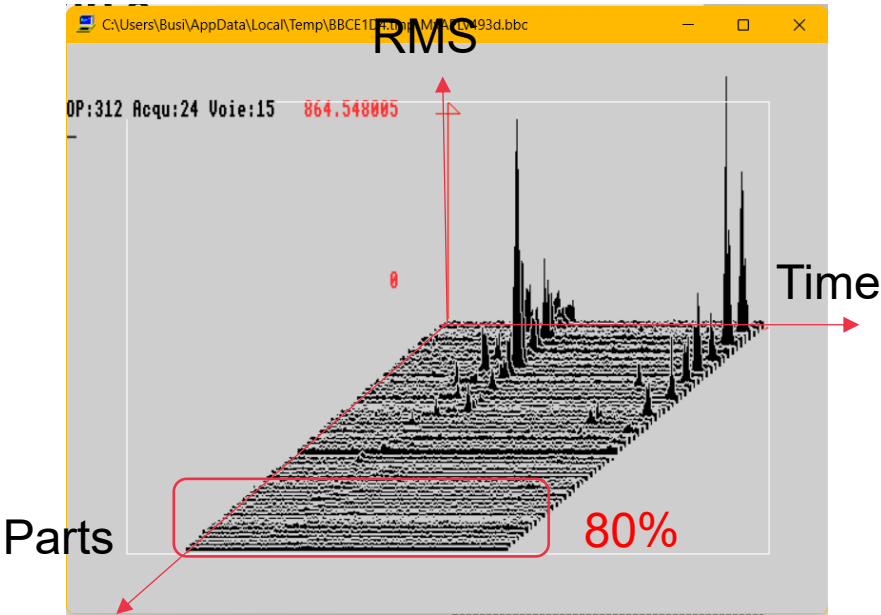
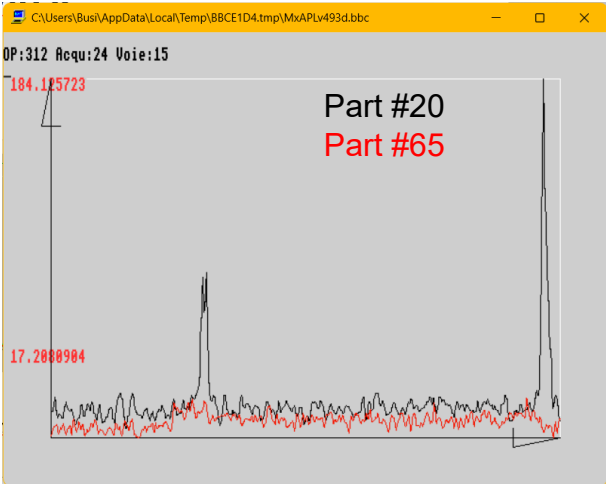


RMS



Before machining

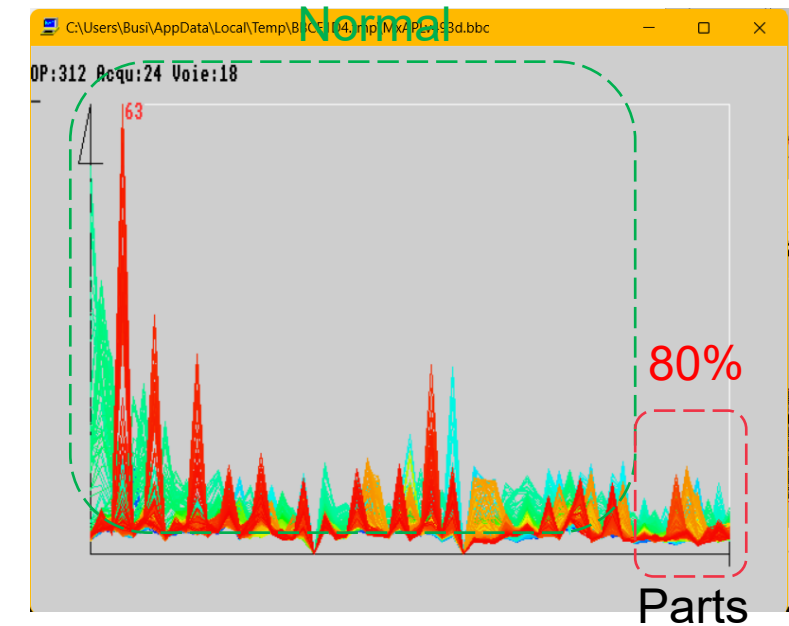
In machining



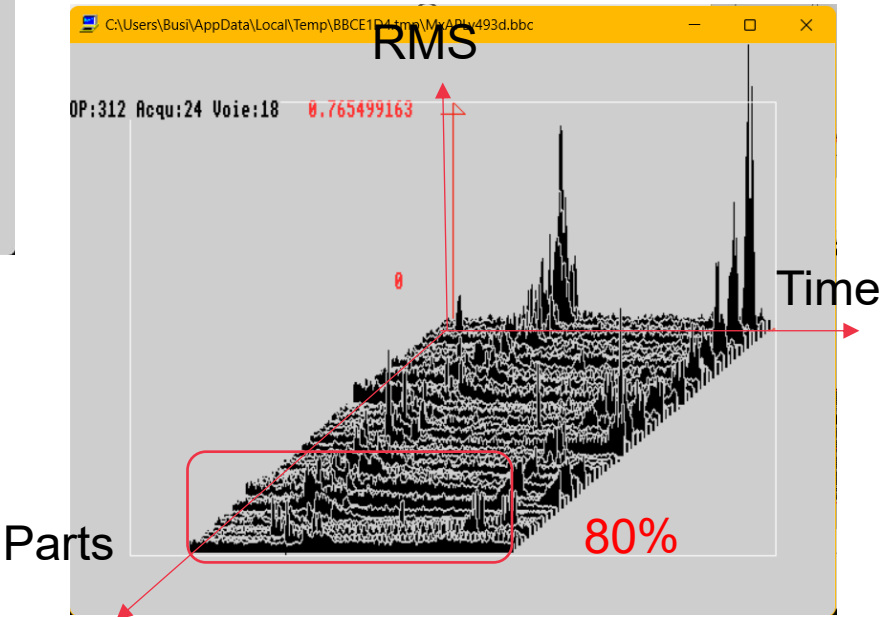
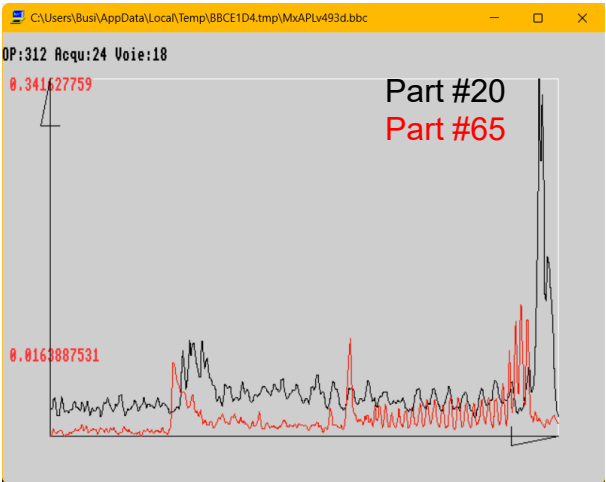
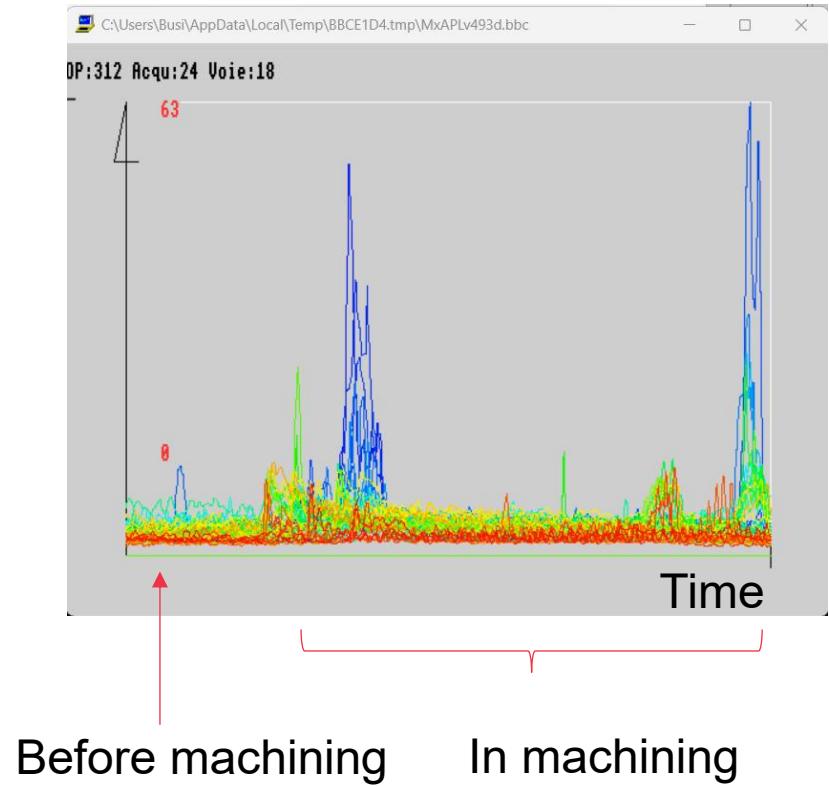
RMS decreases when feedrate decreases

Audio : Vib 11

RMS



RMS



Audio RMS, seems to decrease when feedrate decreases

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="250" description="ALÉSAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="260" description="ALÉSAGE DIAM 11" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="ALÉSAGE DIAM 12" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="280" description="ALÉSAGE DIAM 13" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="290" description="ALÉSAGE DIAM 14" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 313

(last pass #24)

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

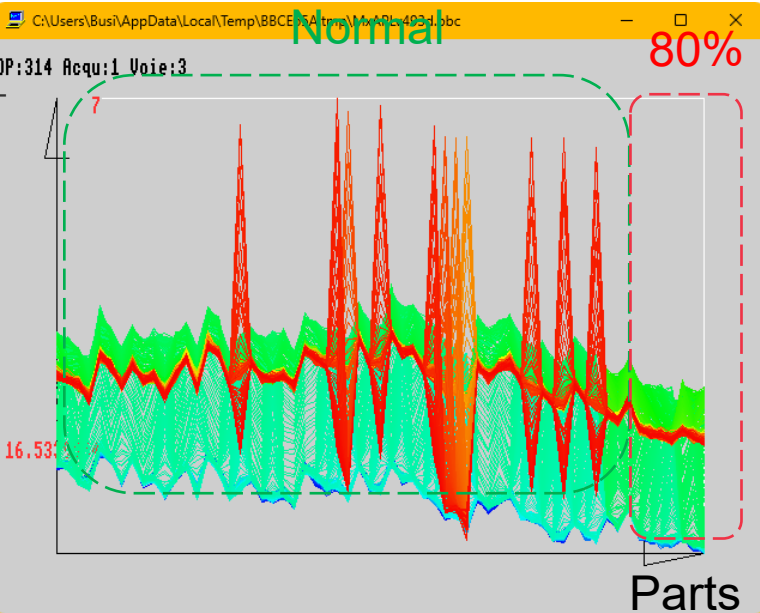
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<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
Le 26/01/26
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

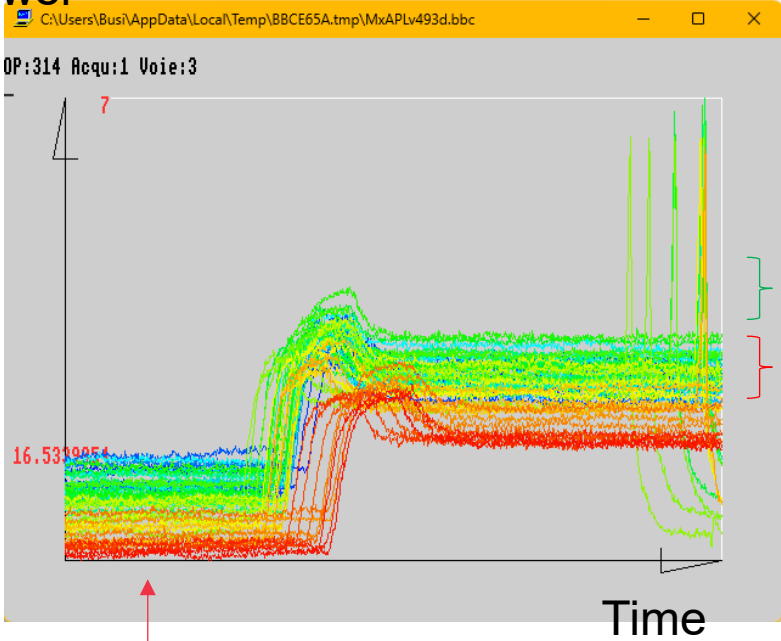
Machining OP 314

Spindle Power : Ai3

Power

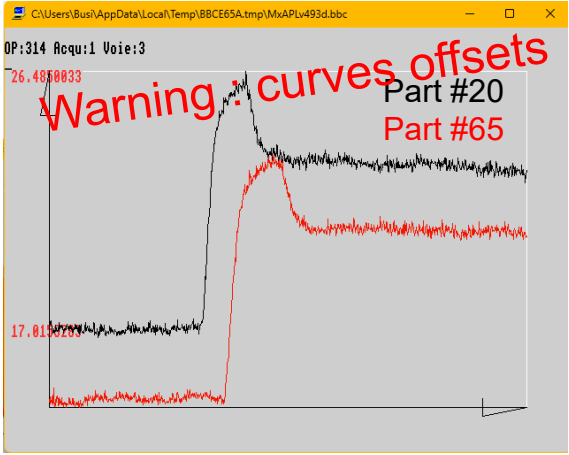


Power

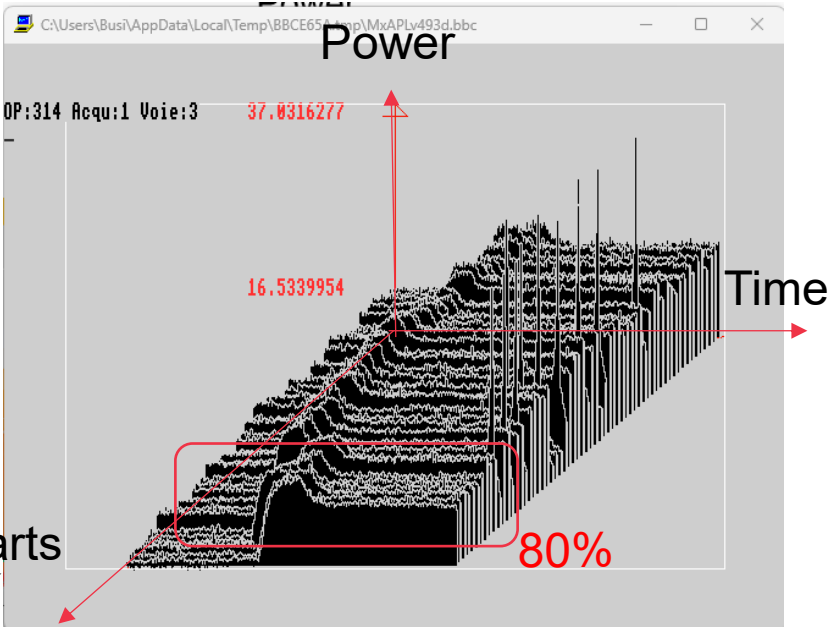


Before machining

In machining



Normal
80%

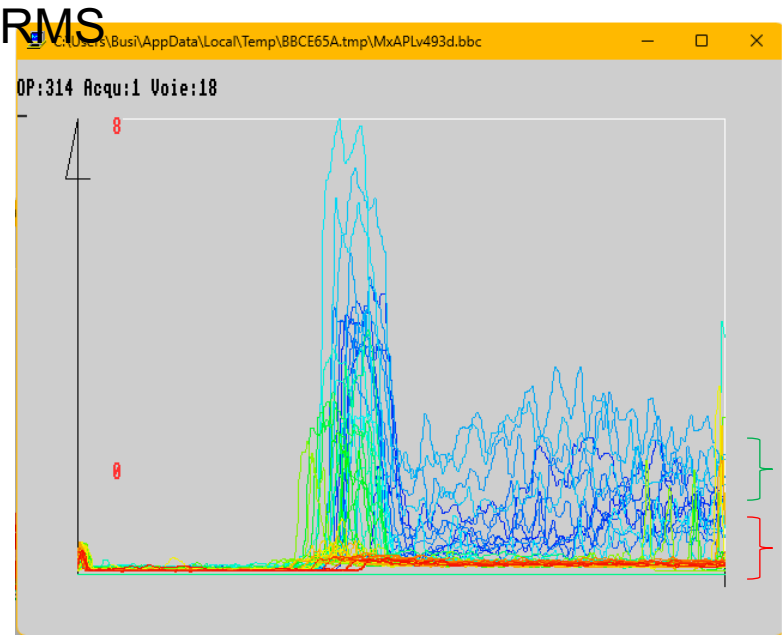
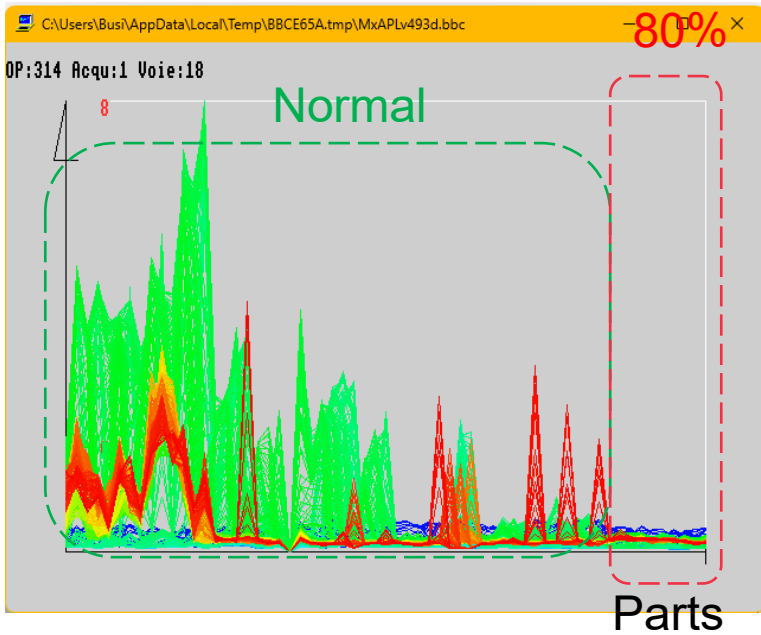


Rem : Temporal shifts sometimes !

Warning to offsets : Spindle power decreases when feedrate decreases => OK

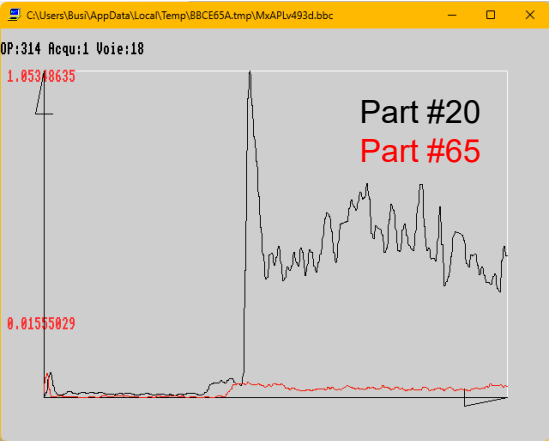
Audio : Vib 11

RMS



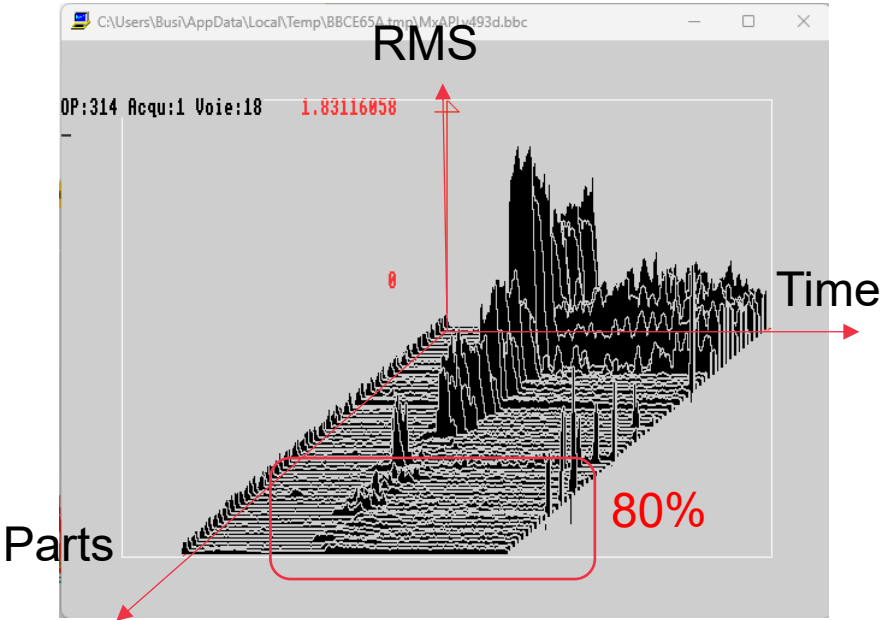
Before machining

In machining



Normal

80%



Audio RMS, seems to decrease when feedrate decreases

Sensors number (for my viewing software) :

- #1 : AI1 (main spindle)
- #2 : AI2 (counter spindle power)
- #3 : AI3 (rotative spindle Power)
- #4 : Fx (roughing turning tool T104)
- #5 : Fy (roughing turning tool T104)
- #6 : Fz (roughing turning tool T104)
- #7 : Fz2 (finishing turning tool T1111)
- #8 : Vib1 (// feedrate on the roughing turning tool T104)
- #9 : Vib2 (// main cutting force on the roughing turning tool T104)
- #10 : Vib3 (boring tool T0202)
- #11 : Vib4 (threading tool T102)
- #12 : Vib5 (cutting-off tool T101)
- #13 : Vib6 (finishing tool T1111)
- #14 : Vib7 (boring tool T1313)
- #15 : Vib8 (milling tool)
- #16 : Vib9 (bar feeder)
- #17 : Vib10 (counter spindle)
- #18 : Vib11 (audio micro)

```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.0999999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 10" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="250" description="AL ESAGE DIAM 10" monitor="true" tool="T02" Color="0;255;0" />
<Operation id="260" description="AL ESAGE DIAM 10" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="PERCAGE DIAM 10" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 315

Sensors number (for my viewing software) :

#1 : AI1 (main spindle)

#2 : AI2 (counter spindle power)

#3 : AI3 (rotative spindle Power)

#4 : Fx (roughing turning tool T104)

#5 : Fy (roughing turning tool T104)

#6 : Fz (roughing turning tool T104)

#7 : Fz2 (finishing turning tool T1111)

#8 : Vib1 (// feedrate on the roughing turning tool T104)

#9 : Vib2 (// main cutting force on the roughing turning tool T104)

#10 : Vib3 (boring tool T0202)

#11 : Vib4 (threading tool T102)

#12 : Vib5 (cutting-off tool T101)

#13 : Vib6 (finishing tool T1111)

#14 : Vib7 (boring tool T1313)

#15 : Vib8 (milling tool)

#16 : Vib9 (bar feeder)

#17 : Vib10 (counter spindle)

#18 : Vib11 (audio micro)

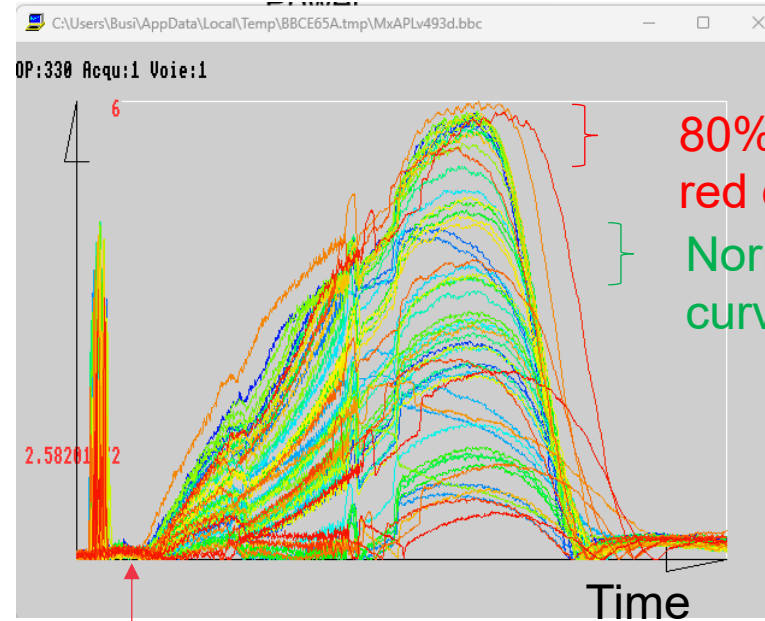
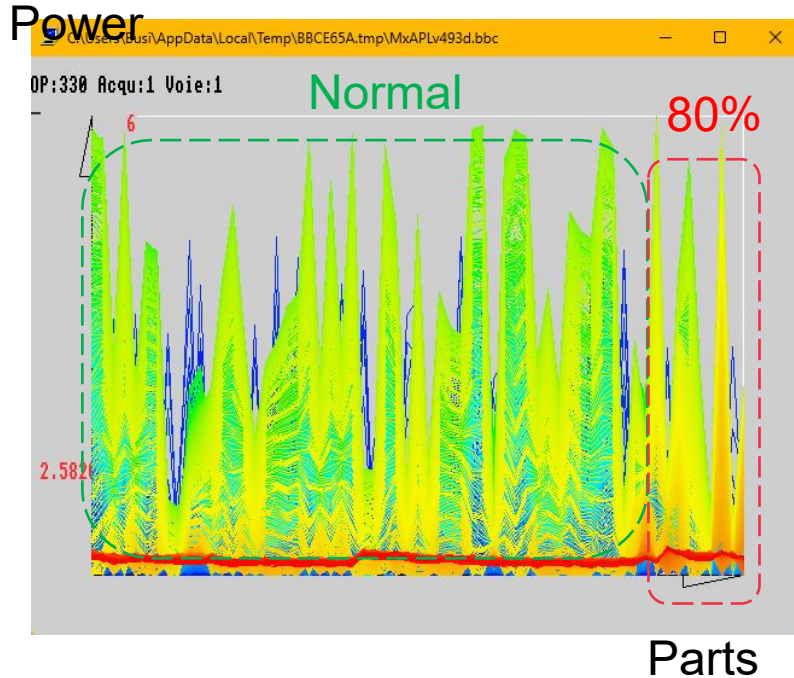
```
<Operation id="110" description="EMBARRAGE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="120" description="EBAUCHE EXTER" monitor="true" tool="T104" Color="0;255;0"
Duration="6.099999999999979" TemplateComputed="true">
<Area name="OutOfWork" start="0.1" end="0.2" reference="true"></Area>
<Area name="InWork" start="1.0" end="2.0" reference="false"></Area>
</Operation>
<Operation id="123" description="NOUVELLE PIECE" monitor="false" tool="T104" Color="0;255;0" />
<Operation id="130" description="FINITION EXTER BP" monitor="true" tool="T104" Color="0;255;0" />
<Operation id="140" description="GORGE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="150" description="FILETAGE" monitor="true" tool="T102" Color="0;255;0" />
<Operation id="160" description="EBAUCHE ARRIERE" monitor="true" tool="T103" Color="0;255;0" />
<Operation id="170" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
<Operation id="210" description="EMBARRAGE" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="220" description="POINTAGE AV" monitor="false" tool="T0505" Color="0;255;0" />
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```

```
<Operation id="230" description="PERCAGE DIAM 8" monitor="true" tool="T0404" Color="0;255;0" />
<Operation id="240" description="PERCAGE DIAM 14" monitor="true" tool="T0303" Color="0;255;0" />
<Operation id="250" description="ALESAGE DIAM 15" monitor="true" tool="T0202" Color="0;255;0" />
<Operation id="260" description="DRESSAGE FACE" monitor="true" tool="T1111" Color="0;255;0" />
<Operation id="270" description="POINTAGE ARR" monitor="false" tool="T1515" Color="0;255;0" />
<Operation id="280" description="PERCAGE DIAM 10" monitor="true" tool="T1414" Color="0;255;0" />
<Operation id="290" description="ALESAGE DIAM 11" monitor="true" tool="T1313" Color="0;255;0" />
<Operation id="300" description="FRAISAGE RENFORT INCLINES (ET SURFACE 1)" monitor="false" tool="T3131"
Color="0;255;0" />
<Operation id="311" description="FRAISAGE SURFACE 1" monitor="false" tool="T3131" Color="0;255;0" />
<Operation id="312" description="FRAISAGE SURFACE 2" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="313" description="FRAISAGE SURFACE 3" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="314" description="FRAISAGE SURFACE 4" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="315" description="FRAISAGE SURFACE 5" monitor="true" tool="T3131" Color="0;255;0" />
<Operation id="320" description="GRAVAGE" monitor="false" tool="T2929" Color="0;255;0" />
<Operation id="330" description="TRONCONNAGE" monitor="true" tool="T101" Color="0;255;0" />
```

Machining OP 330

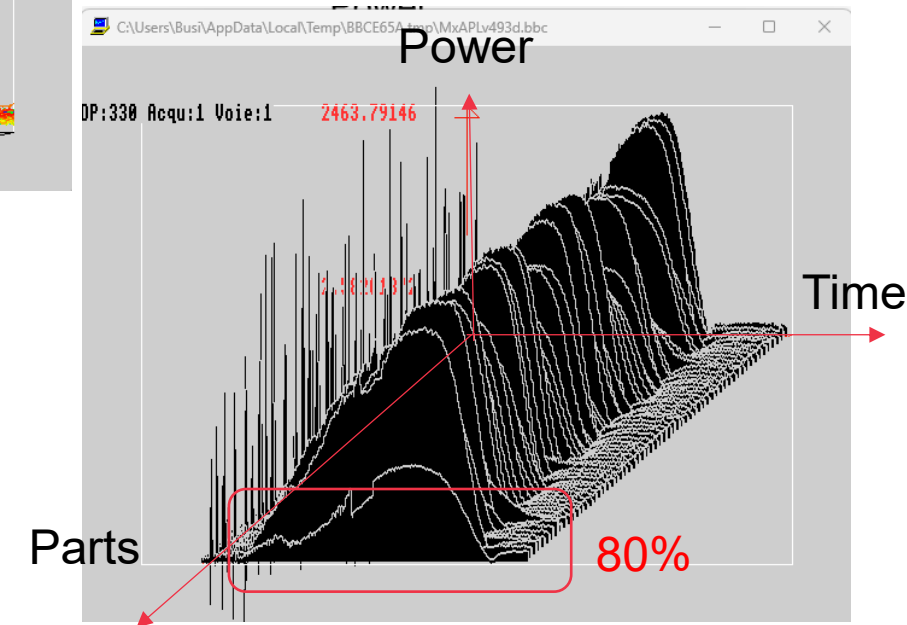
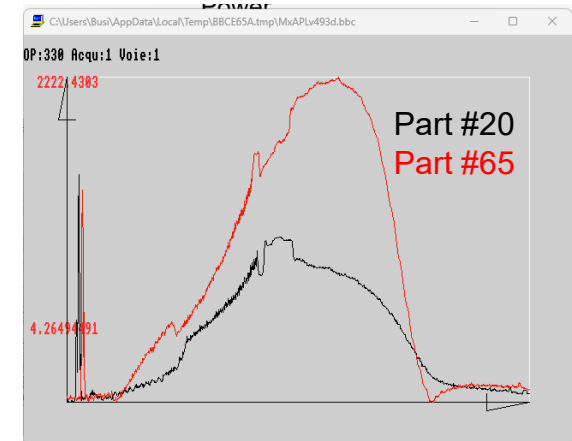
Spindle Power : Ai1

Power



Before machining

In machining

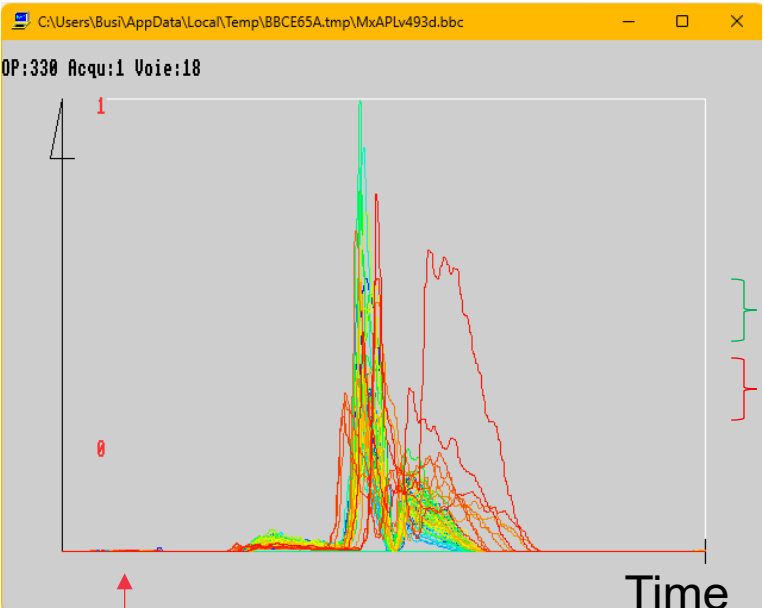
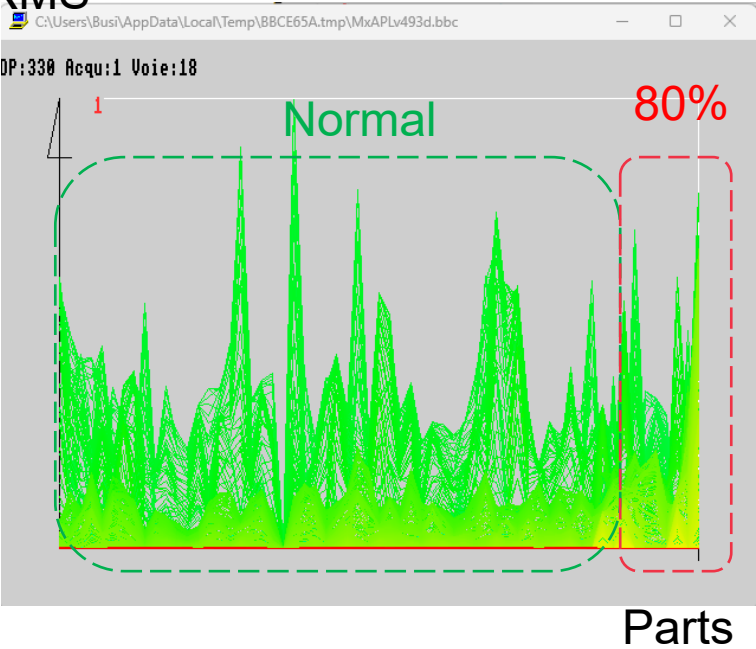


Cutting time increases when feedrate decreases => OK
Spindle power remains the same when feedrate decreases

Audio : Vib 11

RMS

RMS

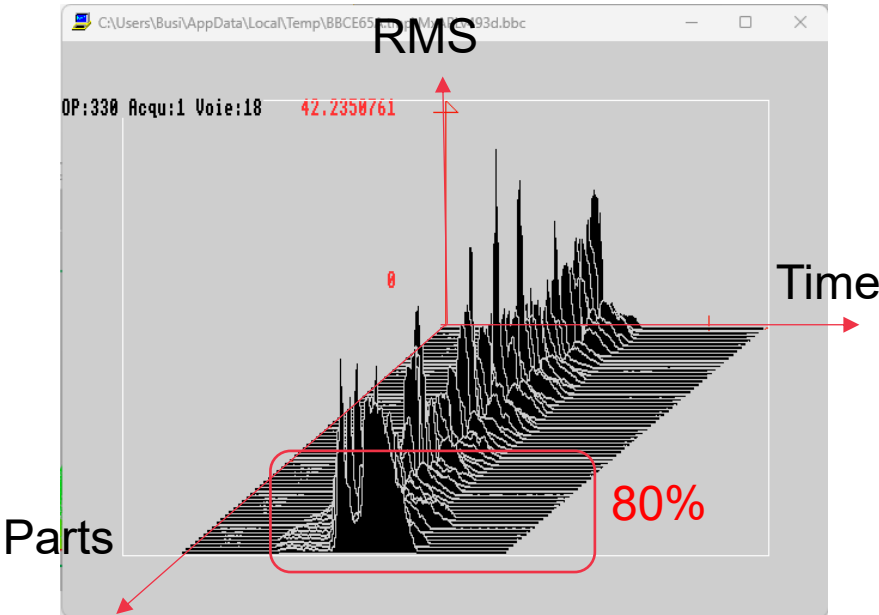
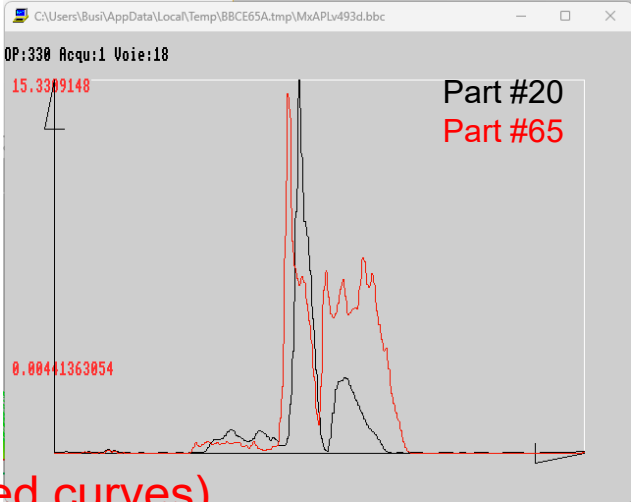


Before machining

In machining

Normal

80% (red curves)



Audio RMS, remains the same when feedrate decreases



Pour un futur industriel
responsable et respectueux
de la planète